

研究

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使用图神经网络结合分子优化增强图特征来改进药物性肝损伤预测

Taeyeub Lee^{1*} 和 Joram M. Posma^{1*}

摘要
目的 药物性肝损伤（DILI）是药物开发中的一个重要问题，常常导致临床试验的中止和药物从市场上的撤回。本研究探讨了图神经网络（GNNs）在DILI预测中的应用，使用分子图表示作为主要输入。
方法 我们评估了几种图神经网络（GNN）架构，包括图卷积网络（GCNs）、图注意力网络（GATs）、图采样与聚合（GraphSAGE）以及图同构网络（GINs），使用了最新的 FDA DILI 数据集和其他分子性质预测数据集。我们提出了一种新方法，该方法通过分子优化创建定制的图数据集，将详细且真实的化学特征（如键长和部分电荷）作为输入纳入 GNN 模型。我们将该模型方法命名为 DILIGeNN。
结果 DILIGeNN 在 DILI 数据集上取得了 0.897 的 AUC，超过了当前 DILI 预测任务的最先进模型。此外，DILIGeNN 在其他基于图的分子预测任务中也优于最先进模型，在 Clintox 数据集上取得了 0.918 的 AUC，在 BBBP 数据集上取得了 0.993 的 AUC，在 BACE 数据集上取得了 0.953 的 AUC，显示出在不同数据集上的强大泛化能力和性能。
结论 DILIGeNN 利用单一图表示作为输入，在结合分子指纹和图结构数据的 DILI 预测中，比最先进的方法表现更好。这些发现突显了我们的分子图生成和 GNN 训练方法的有效性，作为早期药物开发和药物再利用流程的强大工具。
科学贡献：DILIGeNN 是一个 GNN 框架，它从 3D 优化的分子结构中提取图特征，这与靶向药物发现和分子对接模拟中的做法相同。我们的方法是第一个将空间和静电信息编码到单一图表示中的方法，而其他工作通常需要多个图或额外的化学描述符来进行特征表示。我们的方法在训练过程中使用重复早停后的预热启动，与当前最先进的方法相比，在肝毒性（DILI）、渗透性（BBBP）和活性（BACE）预测任务中表现更优。
关键词 药物性肝损伤，特征工程，图神经网络，分子优化

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RESEARCH

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Improving drug-induced liver injury prediction using graph neural networks with augmented graph features from molecular optimisation

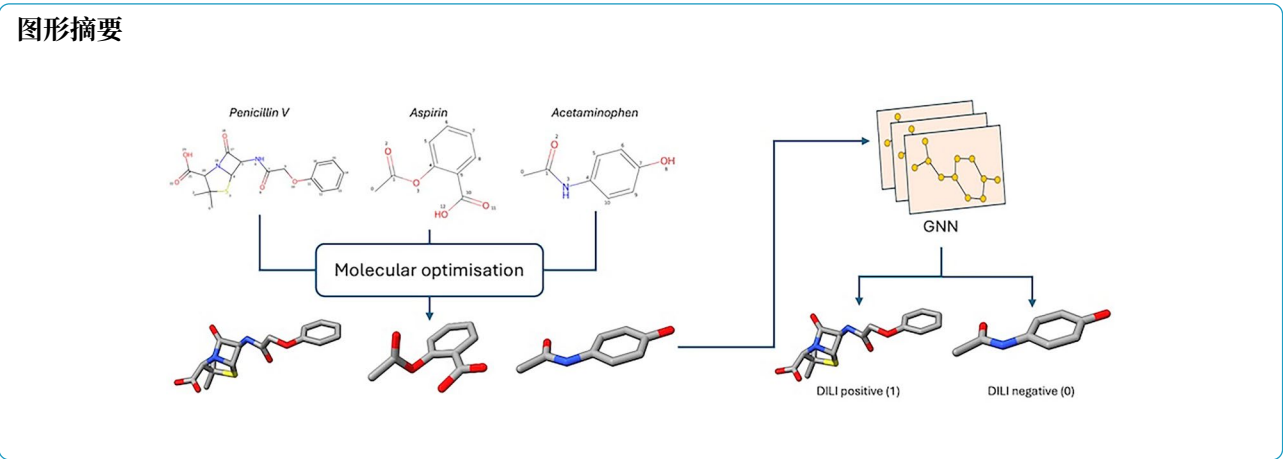
Taeyeub Lee^{1*} and Joram M. Posma^{1*}

Abstract
Purpose Drug-induced liver injury (DILI) is a significant concern in drug development, often leading to the discontinuation of clinical trials and the withdrawal of drugs from the market. This study explores the application of graph neural networks (GNNs) for DILI prediction, using molecular graph representations as the primary input.
Methods We evaluated several GNN architectures, including Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), Graph Sample and Aggregation (GraphSAGE), and Graph Isomorphism Networks (GINs), using the latest FDA DILI dataset and other molecular property prediction datasets. We introduce a novel approach that creates a custom graph dataset, driven by molecular optimisation, that incorporates detailed and realistic chemical features such as bond lengths and partial charges as input into the GNN models. We have named our model approach DILIGeNN.
Results DILIGeNN achieved an AUC of 0.897 on the DILI dataset, surpassing the current state-of-the-art model in the DILI prediction task. Furthermore, DILIGeNN outperformed the state-of-the-art in other graph-based molecular prediction tasks, achieving an AUC of 0.918 on the Clintox dataset, 0.993 on the BBBP dataset, and 0.953 on the BACE dataset, indicating strong generalisation and performance across different datasets.
Conclusion DILIGeNN, utilising a single graph representation as input, outperforms the state-of-the-art methods in DILI prediction that incorporate both molecular fingerprint and graph-structured data. These findings highlight the effectiveness of our molecular graph generation and the GNN training approach as a powerful tool for early-stage drug development and drug repurposing pipeline.
Scientific Contribution: DILIGeNN is a GNN framework that extracts graph features from 3D optimised molecular structures as is done in target-based drug discovery and molecular docking simulation. Our method is the first to encode spatial and electrostatic information into a single graph representation, as opposed to other work that require multiple graphs or additional chemical descriptors for feature representation. Our approach, using warm starts following repeated early stopping during training, outperforms the current state-of-the-art methods in liver toxicity (DILI), permeability (BBBP) and activity (BACE) prediction tasks.
Keywords Drug induced liver injury, Feature engineering, Graph neural networks, Molecular optimisation

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介绍
药物性肝损伤（DILI）是指临床用药意外引起的肝脏损伤，影响肝细胞及其他肝脏细胞。这在药物开发中是一个主要关注点，常常导致临床试验被终止，以及上市药物因对患者安全产生严重影响而被撤回 [1, 2]。DILI约占药物性急性肝衰竭病例的50%，并与超过1000种FDA批准的药物相关 [3, 4]。

药物引起的肝脏毒性大致分为两类：特发性和内在性。内在性DILI通常是剂量依赖性的，并且可以根据药物的药理特性预测，而特发性DILI发生不可预测，并且与剂量无关 [5]。特发性DILI通常有较长的潜伏期，肝毒性在治疗开始数周到数月后才出现 [6]。

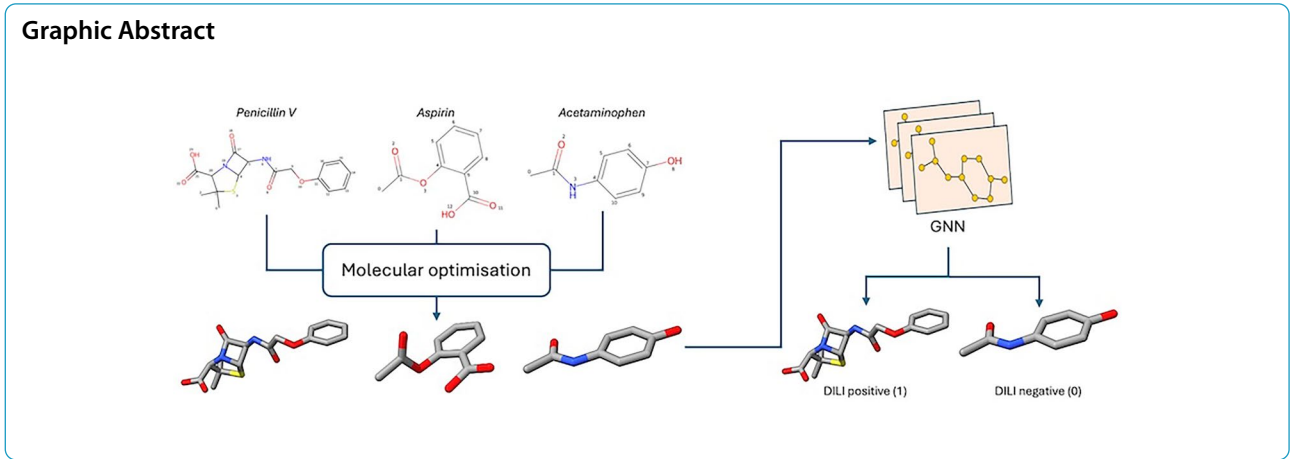
由于动物模型与人类结果的一致性较低，其在早期临床前阶段的检测具有挑战性，在非啮齿动物模型中的一致性率仅为63%，在啮齿动物模型中为43% [7]。此外，尽管体外检测方法已有进展，但约有一半在人体中引起肝脏毒性的药物在临床前毒理学中未被发现 [8]。

因此，非常有必要开发能够准确评估新药候选物肝毒性潜力的计算机预测模型。这类模型能够快速筛选大量药物，帮助在研发早期识别具有高风险的固有性和特异性药物性肝损伤(DILI)药物。这种方法不仅提高了患者的安全性，还降低了临床试验后期失败的风险，并最终使药物研发更加高效和具有成本效益。为了支持准确的计算机DILI预测模型开发，美国食品药品监督管理局(FDA)自 2011 [9–11]起每隔4-5年发布多个版本的DILI数据集。这些更新包括

新的发现、改进的数据质量和扩展的数据覆盖以支持更安全的药物开发，并降低晚期临床试验失败的风险。FDA DILI 数据集的不同版本已被 DILI 预测研究社区广泛使用，用于开发用于 DILI 预测的机器学习（ML）和深度学习（DL）算法。

研究人员使用传统的机器学习 (ML) 和深度学习 (DL) 模型来预测药物引起的肝脏毒性，这在药物开发中构成了重大挑战 [12]。在传统的 ML 方法中，支持向量机 (SVM) 已被用于使用来自 PubChem 指纹的加权分子指纹预测 DILI [13]。类似地，还开发了随机森林 (RF) 模型来预测小鼠肝脏毒性，利用化学描述符和体外 ToxCast 生物活性数据 [14]。这些模型使用了化学描述符和体外 ToxCast 生物活性数据，重点关注包括化学和生物信息的特征 [14]。其他人使用集成学习方法结合多种 ML 算法的预测结果，如线性判别分析、朴素贝叶斯、SVM、分类与回归树以及 k 近邻 (kNN) [15]。这些传统的 ML 方法侧重于整合化学和生物信息

近年来，深度学习（DL）模型在DILI预测任务流程中被越来越多地使用，通常相比传统的机器学习模型表现出更优的性能。下面，我们描述三种使用最新美国FDA DILI数据集（DILIST）[11]的最先进模型，以及两种将FDA DILI数据集与LiverTox数据集[16]，结合使用并在AUC指标下评估的最先进模型，AUC指标是DILI预测研究中广泛认可的标准。



Introduction
Drug-induced liver injury (DILI) refers to the unexpected liver damage caused by clinical drugs, affecting hepatocytes and other liver cells. It is a major concern in drug development, often causing clinical trials to be halted and marketed drugs to be withdrawn due to severe impacts on patient safety [1, 2]. DILI accounts for approximately 50% of drug-induced acute liver failure cases and is linked to more than 1,000 FDA approved drugs [3, 4].

Drug-induced liver toxicity is broadly classified into two types; idiosyncratic and intrinsic. While intrinsic DILI is typically dose-dependent and can be predicted based on the drug's pharmacological properties, idiosyncratic DILI occurs unpredictably and does not correlate with the dosage [5]. The idiosyncratic DILI often have a prolonged latency period, with liver toxicity appearing weeks to months after the initiation of therapy [6].

Its detection at an early pre-clinical stage is challenging due to the low concordance between animal models and human outcomes, with concordance rates of only 63% in non-rodent and 43% in rodent models [7]. In addition, despite the advances in *in vitro* assays, approximately half of the drugs causing liver toxicity in humans are not identified during preclinical toxicology [8].

Therefore, there is a substantial need for developing *in silico* prediction models that can accurately assess the hepatotoxic potential of new drug candidates. Such models would enable the rapid screening of large number of drugs, helping to identify those with high risks of both intrinsic and idiosyncratic DILI early in the development process. This approach not only enhances patient safety but also reduces the risk of late-stage clinical trial failures, and ultimately making drug development more efficient and cost-effective. To support the development of accurate *in silico* DILI prediction models, the US FDA has released several versions of the DILI dataset in 4-5 year intervals since 2011 [9–11]. These updates incorporate

new findings, improved data quality and expand data coverage to support safer drug development and reduces the risk of late-stage clinical trial failures. Different versions of the FDA DILI datasets have been widely used in DILI prediction research community to develop Machine Learning (ML) and Deep Learning (DL) algorithms for DILI prediction .

Researchers use both traditional machine learning (ML) and deep learning (DL) models to predict drug-induced liver toxicity which poses a significant challenge in drug development [12]. In traditional ML approaches, Support Vector Machines (SVMs) have been employed to predict DILI using weighted molecular fingerprints derived from PubChem fingerprints [13]. Similarly, Random Forest (RF) models were also developed to predict mouse liver toxicity, leveraging chemical descriptors and *in vitro* ToxCast bioactivity data [14]. These models used both chemical descriptors and *in vitro* ToxCast bioactivity data, focusing on features that include both chemical and biological information [14]. Others have used ensemble learning approaches to combine predictions from multiple ML algorithms such as linear discriminant analysis, naive Bayes, SVMs, classification and regression trees, and k-nearest neighbours (kNN) [15]. These traditional ML approaches focus on integrating both chemical and biological information due to limitations in predictive performance when relying solely on data from a single domain [12].

In the recent years, deep learning (DL) models are increasingly used in the DILI prediction task pipeline, often yielding superior performance compared to traditional ML models. Below, we describe three state-of-the-art models that used the latest US FDA DILI dataset (DILIST) [11] and two state-of-the-art models that combined the FDA DILI datasets with LiverTox dataset [16], evaluated in AUC metric, which is a widely recognised standard in the DILI prediction research. The

正在考虑的模型包括 DeepDILI [4]、监督子图挖掘 (SSM) [17]、DILIPredictor [18]、深度神经网络 (DNN) [19]、和 DNN-图注意力网络 (DNN-GATNN) [20]、每个模型在 DILI 预测任务流程中都利用了一种或多种先进的机器学习和深度学习技术。DeepDILI模型于2020年发布, 结合了传统的机器学习算法和一个全连接神经网络来预测药物性肝损伤。该模型利用每个机器学习算法的模型级表示, 而不是依赖分子表示, 以有效捕捉与DILI相关的信息。该方法包括逻辑回归、k近邻 (kNN)、支持向量机 (SVM)、随机森林 (RF) 和极端梯度提升 (XGBoost)。这些算法的输出被输入到一个全连接神经网络中, 作为元分类器。该神经网络由多层组成, 使用双曲正切和 Sigmoid等激活函数将输入转换为表示药物引起DILI可能性的概率。DeepDILI取得了0.659的受试者工作特征曲线下面积 (AUC) 和0.687的准确率 (ACC) [4]。

2023 年推出的监督子图挖掘 (SSM) 方法利用随机游走算法提取显式子图特征, 然后用这些特征训练随机森林分类器。该方法通过以监督的方式更新图的转移, 迭代生成用于 DILI 预测的子图, 从而提高子图特征的区分能力。SSM 方法实现了 0.691 的 AUC 和 0.687 的 ACC, 表现优于 DeepDILI [17]。

发表于2024年, DILIPredictor整合了预测的体内和体外数据以预测DILI。该模型使用九个代理DILI标签, 例如线粒体毒性和胆盐外排泵抑制, 以及来自分子指纹的化学结构特征。DILIPredictor是第一个将 FDA DILIst数据集中的化学特征与预测的生物学特征相结合进行DILI预测的模型。该模型实现了0.79的 AUC-PR和0.59 [18]的平衡准确率。尽管该模型获得了相对较低的0.63 AUC-ROC, 但作者采用了基于骨架的拆分, 并专注于与安全性和毒理学更相关的指标, 如阳性似然比和精确度-召回特性, 以强调在实际药物发现过程中对真实有毒化合物的正确识别。

DILI 预测任务中的其他研究将 FDA 发布的 DILI 数据集与额外的 DILI 数据集 LiverTox [16] 结合起来。

据2024年报道, Yang等人提出了一种在不同数据上训练的全连接深度神经网络 (DNN)

分子指纹的类型, 包括基于子结构的 [21, 22]、拓扑 [23, 24], 和环形指纹 [25]。性能最好的深度神经网络使用了环形指纹 (ECFP6), 实现了 AUC 为 0.713 [19]。

在杨等人的工作基础上, Wibowo 等人于2025年提出了一种结合DNN与图注意力神经网络 (DNN-GATNN) 的集成模型。这种方法同时利用了基于子结构的指纹 [21] 和图结构数据, 使模型能够有效捕捉局部子结构以及分子图信息。该集成模型实现了 0.757 的 AUC, 优于之前的最先进模型, 并展示了在 DILI 预测任务中融合基于图的表示的优势 [20]。

图神经网络 (GNNs) 已经成为一类强大的深度学习模型, 能够处理图结构数据, 使其特别适合于分子预测任务。与依赖预定义化学描述符 (如化学指纹或分子理化参数) 的先前模型不同, GNNs 可以直接从分子图中编码的结构信息中学习。这种捕捉复杂关系和局部结构特征的能力, 使其在化学信息学研究中的各种分子性质预测任务中得到了应用。

图卷积网络 (GCN) [26] 已被有效应用于安进的吸收、分布、代谢、排泄 (ADME) 预测服务。通过直接从分子图中学习而不是依赖传统的化学描述符值, GCN显著提高了预测准确性[27]。

在药物-靶点结合亲和力研究中, 图注意力网络 (GAT) 通过使用距离感知注意力机制来结合药物-靶点结合口袋之间的空间信息, 从而提高了预测性能。

通过一种混合方法, 药物的毒性预测得到了改进, 其中图采样与聚合 (GraphSAGE) [30] 用于从分子图中提取结构特征, 而双向门控循环单元 (BiGRU) 则利用 SMILES 字符串提供上下文信息[31]。

在药物-药物相互作用的研究中, 图同构网络 (GIN) [32] 和Node2Vec [33] 的联合应用促进了同构分子图特征的提取, 证明在改善结构相似药物的预测方面是有效的 [34]。

我们的主要目标是探索图神经网络在DILI预测任务中的应用, 同时利用它们从我们自定义设计的数据中直接学习的能力

models under consideration are DeepDILI [4], Supervised Subgraph Mining (SSM) [17], DILIPredictor [18], Deep Neural Network (DNN) [19], and DNN-Graph Attention Network (DNN-GATNN) [20], each leveraging one or more advanced ML and DL techniques in the DILI prediction task pipeline.

Released in 2020, the DeepDILI model integrates conventional machine learning algorithms with a fully connected neural network to predict drug-induced liver injury. This model utilises model-level representations from each ML algorithm, rather than relying on molecular representations, to capture DILI-related information effectively. The approach includes logistic regression, kNN, SVMs, RFs, and extreme gradient boosting. The outputs from these algorithms are fed into a fully connected neural network, which serves as a meta-classifier. This neural network consists of multiple layers, using activation functions such as the hyperbolic tangent and sigmoid to transform the input into probabilities indicating the likelihood of a drug causing DILI. DeepDILI achieved 0.659 Area Under the Receiver Operating Characteristic Curve (AUC) and 0.687 Accuracy (ACC) [4].

Introduced in 2023, the Supervised Subgraph Mining (SSM) approach utilises a random walk algorithm to extract explicit subgraph features, which are then used to train a Random Forest classifier. This method iteratively generates subgraphs for DILI prediction by updating the graph transitions in a supervised manner, increasing the discrimination power of subgraph features. The SSM approach achieved an AUC of 0.691 and an ACC of 0.687, outperforming DeepDILI [17].

Published in 2024, DILIPredictor integrates predicted *in vivo* and *in vitro* data to predict DILI. The model uses nine proxy-DILI labels, such as mitochondrial toxicity and bile salt export pump inhibition, alongside chemical structural features from molecular fingerprints. DILIPredictor is the first model to combine chemical features from the FDA DILIst dataset with the predicted biological features for DILI prediction. The model achieved an AUC-PR of 0.79 and a balanced accuracy of 0.59 [18]. Although the model attained relatively low AUC-ROC of 0.63, the authors employed scaffold-based split and focused on metrics that are more relevant to safety and toxicology, such as positive likelihood ratio and precision-recall characteristics to emphasise correct identification of true toxic compounds in the real-world drug discovery process.

Other studies in DILI prediction task have combined the FDA-released DILI datasets with an additional DILI dataset; LiverTox [16].

Reported in 2024, Yang et al. introduced a fully connected deep neural network (DNN) trained on different

types of molecular fingerprints, including substructure-based [21, 22], topological [23, 24], and circular fingerprints [25]. The best-performing DNN used the circular fingerprints (ECFP6), achieving an AUC of 0.713 [19].

Building upon the work of Yang et al., Wibowo et al. proposed an ensemble model combining a DNN with a graph attention neural network (DNN-GATNN) in 2025. This approach utilised both substructure-based fingerprint [21] and graph-structured data, allowing model to effectively capture local substructures as well as molecular graph information. The ensemble model achieved an AUC of 0.757, outperforming previous state-of-the-art models and demonstrating the advantage of incorporating graph-based representations in the DILI prediction task [20].

Graph neural networks (GNNs) have emerged as a powerful class of DL models capable of processing graph-structured data, making them particularly suitable for molecular prediction tasks. Unlike the previous models that rely on predefined chemical descriptors such as chemical fingerprints or molecular physicochemical parameters, GNNs can directly learn from the structural information encoded within molecular graphs. This ability to capture complex relationships and local structural features has allowed their applications in various molecular property prediction tasks in cheminformatics research.

Graph Convolutional Network (GCN) [26] has been effectively employed in Amgen's absorption, distribution, metabolism, excretion (ADME) prediction service. By learning directly from molecular graphs instead of relying on traditional chemical descriptor values, GCN significantly improved prediction accuracy [27].

In drug-target binding affinity studies, Graph Attention Network (GAT) [28] enhanced the prediction performance by using distance-aware attention mechanisms to incorporate spatial information between drug-target binding pocket [29].

Toxicity prediction for drugs has been improved through a hybrid approach where Graph Sample and Aggregation (GraphSAGE) [30] was used to extract structural features from molecular graphs, while Bi-directional Gated Recurrent Unit (BiGRU) leveraged SMILES strings to provide contextual information [31].

In the study of drug-drug interactions, the combined application of Graph Isomorphism Network (GIN) [32] and Node2Vec [33] facilitated the extraction of isomorphic molecular graph features, proving effective in improving predictions for structurally similar drugs [34].

Our primary objective is to explore the application of GNNs in the DILI prediction task while leveraging their ability to learn directly from our custom-designed

分子图表示。我们的研究受到几个关键观察的激励。

对现有文献的彻底回顾显示, 目前没有模型在 DILI预测中独立使用GNN作为单独的方法。虽然 GNN已在其他分子性质预测任务中显示出成功, 但其在DILI预测中的潜力仍未被探索。这提供了一个机会来研究GNN是否能够提升DILI模型的预测性能。

在近期的最先进模型中, DNN-GATNN 脱颖而出, 成为仅使用化学表示来预测药物性肝损伤 (DILI) 的方法。这种方法很重要, 因为包含额外特征 (如体内或体外数据) 的模型需要广泛的实验设置和专家解读, 从而增加了药物开发流程的复杂性和成本。

此外, 我们将通过使用ClinTox、血脑屏障渗透性 (BBBP) 和 β -分泌酶1 (BACE) 数据集, 在其他分子性质预测任务上测试我们的GNN模型, 以验证其多功能性和泛化能力。这一更广泛的评估旨在评估在DILI预测中表现良好的GNN模型是否也能在其他分子性质预测任务中展现出更高的预测准确性。

对于这些分子预测任务, 一些方法通过结合不同类型的数据和不同的架构, 已经展示了最先进的性能。图拓扑引导的最优传输 (GTOT-tuning) [35] 预训练了一个GIN, 只使用单个分子图 (就像我们这里使用的), 然后针对特定的分子性质预测任务进行微调。其他方法使用额外的数据, 例如带有额外200个化学描述符的分子图, 包括扩展连接指纹 (ECFP) [25], 在定向信息传递神经网络 (D-MPNN) [36], 中, 或者通过几何图嵌入两种分子指纹 (ECFP和使用MACCS键 [21]) 的几何增强分子表示学习 (GEM) [37]。更近期, 使用GIN和GAT模型生成的图嵌入与混合分子指纹一起, 在双图对比学习 (DGCL) 方法 [38], 中被使用, 并通过整合 ECFP与来自图的嵌入。

鉴于这些见解, 本研究旨在评估最突出的GNN架构的有效性: GCN、GAT、GraphSAGE和GIN, 在 DILI及其他分子预测任务中, 仅使用化学表示作为输入。通过仅关注分子

图表示中, 我们希望评估这些 GNN 捕捉相关结构信息的能力, 并在与最新的先进模型相比, 提升 DILI 和其他分子性质预测的性能。

数据与方法

药物性肝损伤数据集

最近发布的药物诱导肝损伤 (DILI) 数据集从美国 FDA官方网站下载 (访问日期: 2024年6月28日)。该数据集是目前可用的最大的二元DILI注释数据集, 包括 768种DILI阳性药物和511种DILI阴性药物, 由 Thakkar等人分类 [11]。分类是通过将DILIranks数据集 [10] 与四个大型文献数据集的数据进行扩展完成的: LiverTox [42], 、Suzuki [43], 、Greene [44], 和 Zhu [45]。这一过程涉及一致性分析和逐步扩展过程, 确保与DILIranks分类结果一致。扩展过程中仅包括符合其各自类别 (DILI阳性或阴性) 75%一致性标准的药物 [11]。数据集包含化合物名称和药物的二元标签, 1: DILI阳性, 0: DILI阴性。SMILES字符串通过 Python (版本3.10) 中使用requests库 (版本2.31) 从PubChem数据库中获取。

其他分子性质预测数据集

MoleculeNet 是一个用于分子性质预测机器学习任务的综合基准套件 [46]。该基准数据集从公共数据集整理而来, 并提供标准化的指标来评估模型性能。我们使用了 MoleculeNet 的三个数据集来评估我们的模型在超出 DILI 数据集的性能。ClinTox 数据集包括 1491 种药物分子, 并包含两个分类任务, 旨在评估模型在 FDA 批准和一般毒性方面的性能。血脑屏障渗透 (BBBP) 数据集源自预测化合物通过血脑屏障渗透性的研究, 这对于开发针对中枢神经系统的药物至关重要。它包含超过 2,053 种化合物的二进制标签, 指示其渗透性特征。BACE 数据集提供与抑制 β -分泌酶 1 (BACE-1) 相关的定量 (IC50) 和定性 (二分类) 数据, 这些数据是

molecular graph representations. Our research is motivated by several key observations.

A thorough review of the existing literature reveals that no current models exclusively utilise GNN as a stand-alone method in DILI prediction. While GNNs have demonstrated success in other molecular property prediction tasks, their potential in DILI prediction remains unexplored. This presents an opportunity to investigate whether the GNNs can enhance the predictive performance of DILI models.

Among the recent state-of-the-art models, DNN-GATNN stands out as the method that predicts DILI using solely chemical representation. This approach is important because models that incorporate additional features such as *in vivo* or *in vitro* data require extensive experimental setups and expert interpretation, increasing the complexity and cost down the drug development pipeline.

Additionally, we will validate the versatility and generalisability of our GNN models by testing them on other molecular property prediction tasks using the ClinTox, Blood-Brain Barrier Permeability (BBBP), and Beta-Secretase 1 (BACE) datasets. This broader evaluation aims to assess whether our GNN models that perform well in DILI prediction can also demonstrate improved predictive accuracy across other molecular property prediction tasks.

For these molecular prediction tasks, several methods have demonstrated state-of-the-art performance by utilising different types of data in conjunction with different architectures. Graph topology induced optimal transport (GTOT-tuning) [35] pretrained a GIN using only a single molecular graph (as we are using here) followed by fine-tuning for specific molecular property prediction tasks. Other methods use additional data such as molecular graphs with additional 200 chemical descriptors, including extended-connectivity fingerprint (ECFP) [25], in a directed message passing neural network (D-MPNN) [36], or by geometric graph embedding of two molecular fingerprints (ECFP and using MACCS keys [21]) using geometry-enhanced molecular representation learning (GEM) [37]. More recently, graph embeddings generated from GIN and GAT models with mixed molecular fingerprints were used in a dual-graph contrastive learning (DGCL) approach [38], and by integrating ECFP with embeddings from a graph-based model [39] and the MoLFormer language model [40, 41].

Given these insights, this study aims to evaluate the effectiveness of the most prominent GNN architectures: GCN, GAT, GraphSAGE and GIN, in DILI and the other molecular prediction tasks using only chemical representations as input. By focusing solely on the molecular

graph representations, we seek to assess the capability of these GNNs capturing relevant structural information and improve DILI and the other molecular property prediction performance compared to the recent state-of-the-art models.

Data and methods

Drug induced liver injury dataset

The most recent release of the drug-induced liver injury (DILI) dataset was downloaded from [the US FDA official website](#) (accessed on 28/06/2024). The dataset is the largest binary DILI annotation dataset available which composes of 768 DILI positive and 511 DILI negative drugs classified by Thakkar et al. [11]. The classification was performed by augmenting the DILIranks dataset [10] with data from four large literature datasets: LiverTox [42], Suzuki [43], Greene [44], and Zhu [45]. This involved a concordance analysis and incremental augmentation process, ensuring consistency with the DILIranks classifications. The augmentation process only included drugs that met a 75% concordance criterion for their respective categories: DILI positive or negative [11]. The dataset contains compound names and binary labels of drugs, 1: DILI positive and 0: DILI negative. SMILES strings were retrieved from the PubChem database using the requests library (version 2.31) in Python (version 3.10).

Other molecular property prediction datasets

MoleculeNet is a comprehensive benchmark suite for machine learning tasks in molecular property prediction [46]. This benchmark dataset is curated from public datasets and provides standardised metrics to assess model performance. We utilised three datasets from MoleculeNet to evaluate our model performance beyond the DILI dataset. The ClinTox dataset includes 1491 drug molecules and features two classification tasks aimed at evaluating model performance on FDA approval and general toxicity. The Blood–Brain Barrier Penetration (BBBP) dataset is derived from research focused on predicting the permeability of compounds through the blood-brain barrier, which is crucial for developing drugs targeting the central nervous system. It contains binary labels for over 2,053 compounds indicating their permeability properties. The BACE dataset provides both quantitative (IC50) and qualitative (binary classification) data related to inhibition of the enzyme beta-secretase 1 (BACE-1), which is implicated in Alzheimer’s disease. It consists of 1,522 compounds, and we utilised the qualitative data for our analysis. We obtained the chemical structures for these datasets using the provided SMILES strings. The dataset was downloaded using PyTorch

Geometric (版本 2.5.3, 访问日期 2024 年 4 月 15 日) [47]。

药物数据预处理

我们创建了一个内部的分子预处理和标准化算法, 以确保 DILI 数据集中分子表示被清理、标准化和验证, 以便进一步分析和在化学信息学应用中使用。该过程的目标是将潜在多样且非标准化的 SMILES 字符串转换为一致的格式, 这对于可靠的下游分析 (如包括本研究在内的结构毒性关系研究) 至关重要。在整个预处理步骤中, 无法预处理的药物分子会被记录并附上相应的错误信息保存下来。用于预处理的数据可在支持文档中的文件 DILIst standardised.csv 中获取。

化学元素筛选

该过程首先通过迭代DILI数据集中的每个SMILES字符串开始。该算法尝试将这些字符串转换为RDKit分子对象 (RDKit版本2024.03.1)。对于有效的分子, 算法会遍历每个分子的化学元素, 以确保分子仅包含通常用于有机化学和药物合成的10种常见元素。这些元素包括氢、碳、氮、氧、氟、磷、硫、氯、溴和碘, 这些元素通常存在于具有期望药代动力学和药效学特性的稳定药物样分子中。此步骤对于过滤掉含有不常见元素的药物至关重要, 这些元素在FDA批准的药物中较少出现, 从而排除了可能导致DILI预测模型训练偏差的含有非典型元素的特殊药物 (例如含砷的乙砒唑和含硼的硼替佐米)。我们已排除了34种具有

忽略不太相关的片段, 如盐类和溶剂。在对母分子进行标准化和鉴定后, 算法枚举所有可能的互变异构体以标准化互变异构形式。通过识别单一的互变异构体, 这确保了分子在不同互变异构状态之间的一致性。这一步对于那些可以存在多种互变异构形式的化合物尤其重要, 例如华法林 [49]。

分子图生成 分子优化

每种药物的 SMILES 字符串用于生成 RDKit mol 格式的二维分子结构, 然后添加氢原子, 以确保每个结构的氢原子都被明确表示, 这是分子模拟和优化中必不可少的。接着, 使用 RDKit 的 EmbedMolecule 函数实现的距离几何方法生成药物分子的初始三维构象。生成过程涉及根据其分子结构中的化学键、角度和扭转产生的距离约束来创建原子的空间排列。在三维构象生成过程中, 允许最多 5000 次尝试, 从随机原子坐标开始, 以生成满足所有几何约束的有效结构。构象生成仅确保生成的构象在合理范围内, 并遵循基本几何约束, 因此我们进一步对构象进行优化。

创建自定义图表数据集

PyTorch Geometric 的 MoleculeNet 数据集被广泛用于分子性质预测的机器学习任务。这些数据集将分子表示为图, 其中节点对应于原子, 边表示化学键。节点特征通常包括原子序数、度、杂化类型和芳香性 [47]。PyTorch 的 MoleculeNet 数据集 [46] 常用于基于图的算法, 例如图神经网络, 这些算法利用这些结构表示来分析原子之间的相互作用以进行分子性质预测任务。然而, 这些数据集不包含三维信息。

Geometric (version 2.5.3, accessed on 15/04/2024) [47]。

Drug data pre-processing

We created an in-house molecule pre-processing and standardisation algorithm to ensure that molecular representations in the DILI dataset is cleaned, normalised and validated for further analysis and use in cheminformatics applications. The goal of this process is to convert potentially diverse and non-standardised SMILES strings into a consistent format, which is crucial for reliable downstream analysis such as structure toxicity relationship studies including current study. Throughout the pre-processing steps, drug molecules that could not be pre-processed were recorded with appropriate error messages and saved. The data used for pre-processing is available in the file DILIst_standardised.csv in the supporting documents.

Chemical element screening

The process begins by iterating through each SMILES string in the DILI dataset. The algorithm attempts to convert these strings into molecular objects in RDKit mol formats (RDKit version 2024.03.1). For valid molecules, the algorithm iterates through the chemical elements of each molecule to ensure that the molecule only contains 10 common elements typically used in organic chemistry and drug synthesis. These elements include Hydrogen, Carbon, Nitrogen, Oxygen, Fluorine, Phosphorus, Sulphur, Chlorine, Bromine and Iodine, and these elements are commonly found in stable, drug-like molecules with the desired pharmacokinetic and pharmacodynamic properties. This step is vital for filtering out drugs with uncommon elements which are less frequent in FDA-approved drugs [48], thereby excluding specialised drugs with uncommon elements (such as Acetarsol with Arsenic, and Bortezomib with Boron) which may cause bias in training the DILI prediction models. We have excluded 34 Drugs with the uncommon elements which can be found in our [GitHub](#).

Molecule standardisation

Once a molecule passes the chemical element screening, the molecular standardisation process begins. This involves molecular normalisation steps such as removing explicit hydrogen atoms, disconnecting metal atoms, and neutralising ionic charges to produce a standardised molecular representation. A critical part of this process is identifying the parent molecule, especially in structures with multiple fragments, which is often encountered in medicinal chemistry. The parent molecule is the largest fragment, and its identification focuses the analysis on the active structure of interest, allowing smaller,

less relevant fragments such as salts and solvents to be ignored. After normalisation and identification of the parent molecule, the algorithm enumerates all possible tautomers to standardise tautomeric forms. This ensures consistency across different tautomeric states of a molecule by identifying a single tautomeric form. This step is particularly important for compounds that can exist in multiple tautomeric forms, such as Warfarin [49].

Molecular graph generation

Molecule optimisation

The SMILES string of each drug was used to generate a 2D molecular structure in RDKit mol format, after which hydrogen atoms were added to ensure each structure has all its hydrogen atoms explicitly represented which is essential in molecular simulations and optimisations. Then, An initial 3D conformation of the drug molecule was generated using distance geometry methods as implemented in RDKit's EmbedMolecule function [23]. The generation process involved creating a spatial arrangement of atoms based on distance constraints derived from the chemical bonds, angles, and torsions by its molecular structure. During the 3D conformation generation, a maximum of 5,000 attempts was allowed, starting from random atomic coordinates to produce a valid structure that satisfied all geometric constraints [50]. The conformer generation only ensures that generated conformation is reasonably plausible and follows the basic geometric constraints, therefore, we further optimised the conformer using Merck Molecular Force Field (MMFF) method, as implemented in RDKit's MMFFOptimizeMolecule function [23]. MMFF takes various intra-molecular forces into account including bond stretching, angle bending, torsions, van der Waals interactions, and electrostatics, to generate molecular geometries that are consistent with experimental data by adjusting the positions of atoms to minimise the overall molecular energy to find the most stable conformation [51].

Creation of custom graph dataset

The PyTorch Geometric MoleculeNet datasets are widely used for machine learning tasks in molecular property prediction. These datasets represent molecules as graphs, where nodes correspond to atoms and edges represent bonds. Node features typically include atomic number, degree, hybridisation, and aromaticity [47]. PyTorch MoleculeNet datasets [46] are commonly used in graph-based algorithms, such as graph neural networks, which utilise these structural representations to analyse interactions between atoms for molecular property prediction tasks. However, these datasets do not include 3D

表 1 PyTorch Geometric MoleculeNet 数据集和我们自定义图数据集中包含的节点和边特征比较

功能类型	PyTorch 几何分子网	Our 自定义图表
节点特征		
原子序数	✓	✓
原子度	✓	✓
形式电荷	✓	✓
杂交	✓	✓
芳香性	✓	✓
氢原子数量	✓	✓
在环中	✓	✓
手性	✓	✓
自由基电子		✓
Gasteiger部分电荷		✓
总化合价		✓
显式化合价		✓
边缘特征		
键类型	✓	✓
在环中	✓	✓
键长		✓
自定义图数据集包含额外的特征, 例如 Gasteiger 部分电荷、总价电子数以及来源于三维优化构象的键长		

构象数据, 这限制了它们直接捕捉分子空间特征的能力。

相比之下, 我们的自定义图数据集 (表1) 包含分子优化, 提供3D空间信息, 以更准确地描述分子几何结构, 从中使用RDKit的rdMolTransforms.GetBondLength函数计算的键长作为边特征[23], , 该函数基于3D坐标计算键合原子之间的欧几里得距离。虽然这些键长并不代表生物环境中的真实分子构象, 但这种方法提供了接近实验数据的更现实近似, 而不是依赖经验表中的平均键长[52, 53]。这种方法是药物化学领域中分子对接模拟的标准程序, 因为准确的几何结构至关重要, 它影响配体与靶点结合位点内的分子稳定性[54], , 但在人工智能社区尚未被广泛采用。

此外, 本研究创建的自定义数据集增加了诸如Gasteiger部分电荷、总价电子以及明确价电子等特性。Gasteiger电荷是使用RDKit的ComputeGasteigerCharges函数计算的, 而价电子属性则通过RDKit的原子对象方法获得。

GetTotalValence 和 GetExplicitValence [23]。这些附加特性提供了关于分子内电子分布的详细信息, 通过提供对分子结构的更详细表示, 支持对复杂分子性质的更准确预测 [56]。

键长变化分析和统计测试
键长变化是通过每个分子分子优化前后的百分比差异来评估的。为了总结数据, 我们计算了基本的统计指标, 包括均值、中位数、标准差和范围, 针对标准化和非标准化的 SMILES 表示, 使用 Pandas 库 (版本 2.2.2) [57] 的 describe 函数。为了统计评估标准化和非标准化 SMILES 间键长变化的差异, 我们采用了配对 t 检验。该检验适用于比较两个相关样本, 具体而言是同一组分子在标准化前后的键长变化。配对 t 检验使用 Scikit-learn 库 (版本 1.3.1) [58] 中的 ttest_rel 函数进行。该方法使我们能够确定标准化过程是否会在键长变化上产生具有统计学显著性的差异。

基本分子图表示的创建
在我们的特征比较研究中, 我们利用了一个基础的分子图表示作为建模分子结构的基础方法。这种表示将每个分子简化为一个图, 其中原子被表示为节点, 原子之间的键被表示为边。在这种基础的图表示中, 主要特征包括节点的原子序数和边的键类型。这些特征提供了关于分子结构的基本信息, 捕捉了进行初步性质预测任务所需的基本方面。许多基于图的预测方法, 例如SSM模型, 利用这种基础的分子图作为输入特征 [17]。

深度图神经网络的训练与评估
在这项研究中, 我们开发了深度图神经网络 (GNNs), 以使用我们自定义设计的图数据集执行基于图的学习任务, 该数据集包括节点特征、边特征以及用于药物性肝损伤 (DILI) 二分类的目标标签。本研究使用的模型包括图卷积网络 (GCNs)、图注意力网络 (GATs)、GraphSAGE 和图同构网络 (GINs)。选择这些架构是为了准确捕捉

Table 1 Comparison of node and edge features included in the PyTorch Geometric MoleculeNet dataset and our custom graph dataset

Feature type	PyTorch geometric moleculeNet	Our custom graphs
Node features		
Atomic number	✓	✓
Atom degree	✓	✓
Formal charge	✓	✓
Hybridisation	✓	✓
Aromaticity	✓	✓
Number of hydrogens	✓	✓
In ring	✓	✓
Chirality	✓	✓
Radical electrons		✓
Gasteiger partial charges		✓
Total valence		✓
Explicit valence		✓
Edge features		
Bond type	✓	✓
In ring	✓	✓
Bond length		✓
The custom graph dataset incorporates additional features such as Gasteiger partial charges, total valence, and bond length derived from 3D optimised conformers		

conformational data, which limits their ability to capture spatial molecular features directly.

In contrast, our custom graph dataset (Table 1) incorporates molecular optimisation, providing 3D spatial information for a more accurate depiction of molecular geometry, from which bond lengths are calculated as edge features using RDKit’s rdMolTransforms.GetBondLength function [23], which computes the Euclidean distance between bonded atoms based on their 3D coordinates. While these bond lengths do not represent the true molecular conformation in a biological environment, this approach offers a more realistic approximation close to experimental data, rather than relying on average bond lengths from empirical tables [52, 53]. This approach is a standard procedure in the field of medicinal chemistry for molecular docking simulations where accurate geometry is crucial as it influences molecular stability within ligand-target binding sites [54], but it has not been widely adopted in the AI community.

Furthermore, our custom dataset created in this study adds features such as Gasteiger partial charges [55], total valence, and explicit valence. Gasteiger charges were computed using RDKit’s ComputeGasteigerCharges function, while valence properties were obtained via RDKit’s atom object methods

GetTotalValence and GetExplicitValence [23]. These additional features provide detailed information about the electronic distributions within the molecule supporting more accurate predictions of complex molecular properties by offering a more detailed representation of molecular structure [56].

Bond length change analysis and statistical testing
The bond length changes were evaluated as a percentage difference before and after molecular optimisation for each molecule. To summarise the data, we computed basic statistical metrics, including the mean, median, standard deviation, and range, for both standardised and non-standardised SMILES representations using the describe function from the Pandas library (version 2.2.2) [57]. To statistically assess the differences in bond length changes between standardised and non-standardised SMILES, we employed a paired t-test. This test is suitable for comparing two related samples, specifically the bond length changes for the same set of molecules before and after standardisation. The paired t-test was conducted using the ttest_rel function from the from Scikit-learn library (version 1.3.1) [58]. This approach allows us to determine whether the standardisation process results in statistically significant differences in bond length changes.

Creation of basic molecular graph representation
In our feature comparison study, we utilised a basic molecular graph representation as a foundational approach to model molecular structures. This representation simplifies each molecule into a graph, where atoms are depicted as nodes and bonds between them as edges. In this basic graph representation, the primary features include the atomic number for nodes and the bond type for edges. These features provide basic information about the molecular structure, capturing fundamental aspects necessary for initial property prediction tasks. Many graph-based prediction approaches such as the SSM model utilised this basic molecular graph as input features [17].

Training and evaluation of deep graph neural networks
In this study, we developed deep graph neural networks (GNNs) to perform graph-based learning tasks using our custom-designed graph dataset, which includes node features, edge features, and target labels for the binary classification of drug-induced liver injury (DILI). The models used in this study include Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), GraphSAGE, and Graph Isomorphism Networks (GINs). These architectures were chosen to accurately capture the

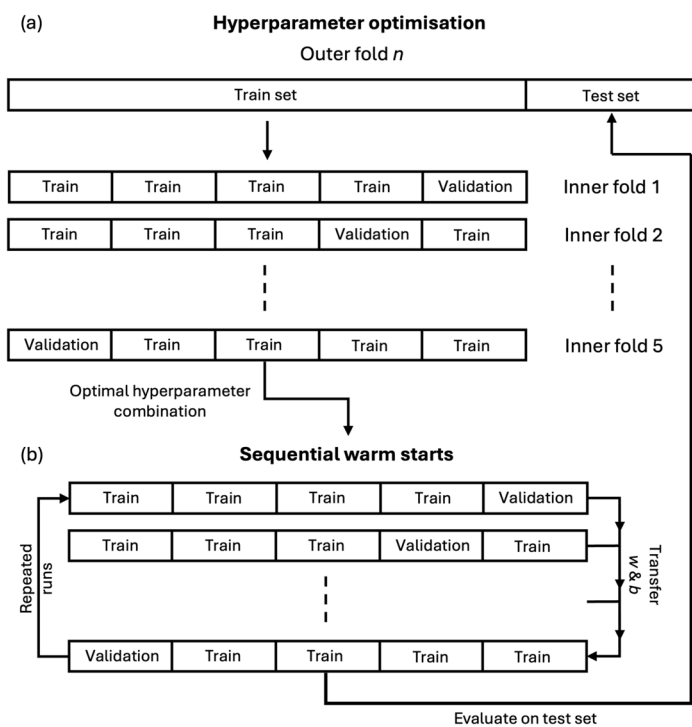


图1 超参数优化和顺序热启动的嵌套交叉验证框架。本图展示了在嵌套交叉验证框架中进行超参数优化、模型训练和评估的顺序热启动工作流程。a 超参数优化, 通过五个内折找到最优超参数。b 顺序热启动, 在内循环中多次运行以微调模型参数。随后, 在外折测试集上评估训练好的模型。

分子图内部的关系, 从而提高分子性质预测的准确性。为了确保这些模型的稳健训练、优化和评估, 我们采用了嵌套交叉验证 (CV) 框架进行超参数优化、模型重新初始化以及顺序预热策略。这些方法旨在最大化不同随机初始化下的性能稳定性, 并提高我们模型的泛化能力。此外, 我们在其他二元分子性质预测数据集上进行了相同的实验, 包括来自MoleculeNet数据集的Clintox、BBBP和BACE, 以评估我们的模型在不同分子性质预测任务中的性能, 并验证我们的训练方法的有效性。随后的章节将详细介绍实验设置, 包括模型架构、训练流程以及用于衡量模型性能的评估指标。

测试集 (占数据的25%), 而剩下的三折则进一步分成五个分层折用于超参数调优的内部循环。使用这种分割过程是为了确保用于评估模型的测试分割 (见下节关于指标的说明), 在超参数优化后仍然保持独立。此外, 在内部循环中执行交叉验证 (CV) 可以使用验证集找到最优超参数, 并防止过拟合 (从而使模型更具泛化性)。在内部循环的每次迭代中, 一折被用作超参数优化的验证集 (占15%), 其余四折用作内部折训练集 (占60%)。使用135种超参数组合对模型进行训练和评估, 包括批量大小、隐藏通道数量、丢弃率和学习率, 这些组合列在补充表1中。每个外部折的最优超参数组合基于综合排名选择。

用于超参数优化的嵌套交叉验证

图1a显示了用于外层折叠超参数优化的嵌套交叉验证框架。对于每个GNN模型, 数据最初被分为四个分层的外层折叠。在外层折叠的每次迭代中, 一个折叠用作

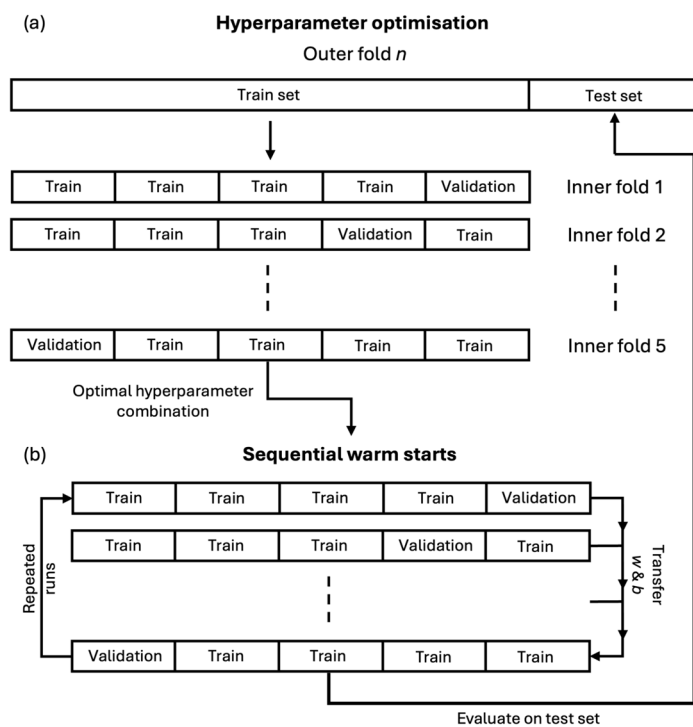


Fig. 1 Nested Cross-Validation Framework for Hyperparameter Optimisation and Sequential Warm Starts. This figure presents a workflow of hyperparameter optimisation, sequential warm starts for model training and evaluation in the nested cross-validation framework. **a** Hyperparameter Optimisation to find the optimal hyperparameter across five inner folds. **b** Sequential warm starts for fine-tuning the model parameters within the inner loop for multiple runs. Subsequently, the trained model was evaluated on the outer fold test set

relationships within molecular graphs, thereby improving the predictive accuracy of molecular property prediction. To ensure robust training, optimisation and evaluation of these models, we employed a nested cross-validation (CV) framework for hyperparameter optimisation, model reinitialisation and sequential warm starts strategies. These methods aimed to maximise performance stability across different random initialisations and improve the generalisability of our models. Additionally, we conducted the same experiment on other binary molecular property prediction datasets, including Clintox, BBBP, and BACE, sourced from the MoleculeNet dataset [46] to evaluate our model performance across different molecular property prediction tasks and validate the effectiveness of our training approach. The subsequent sections provide a detailed account of the experimental setup, including model architectures, training procedures, and the evaluation metrics used to measure model performance in different stages of our nested CV framework.

Nested cross-validation for hyperparameter optimisation

Figure 1a shows the nested CV framework for outer fold hyperparameter optimisation. For each GNN model, the data was initially divided into four stratified outer folds. In each iteration of the outer folds, one fold served as

the test set (25% of the data), while the remaining three folds were further split into five stratified folds for the inner loop for hyperparameter tuning. This process of splitting was used to ensure the test splits used for evaluation (see section below on metrics) of the models with hyperparameters optimised remain independent. Moreover, performing CV in the inner loop allows the optimal hyperparameters to be found using a validation split and prevents overfitting (thereby making it more generalisable). In each iteration of this inner loop, one fold was used as the validation set (15%) for hyperparameter optimisation and the remaining four folds were used as the inner fold training set (60%). The model was trained and evaluated using 135 hyperparameter combinations including batch size, number of hidden channels, dropout rate and learning rate, listed in Supplementary Table 1. The optimal hyperparameter combination for each outer fold was selected based on a combined ranking of the highest median and minimum validation AUC scores across the inner folds. This approach ensured that the selected hyperparameters performed consistently well across different data splits. This entire process was repeated across all outer folds ($n=4$). For data splitting we used stratified cross-validation algorithm from `scikit-learn` (version 1.3.1).

模型重新初始化和顺序热启动以进行微调

在确定最优超参数后, 我们通过使用 PyTorch (版本 2.3.1) [59] 的 20 个不同随机种子训练模型, 解决了随机权重初始化引入的变异性。对于每个外层折叠, 我们根据从内层折叠获得的中位数和最小验证 AUC 分数的综合排名选择表现最好的种子。该模型重新初始化步骤旨在提高所有内层折叠的性能稳定性, 并确保后续工作的可重复性。为了进一步优化模型, 我们采用了顺序热启动方法对参数进行微调 (见图 1b)。顺序热启动方法中的每次运行都包括在所有内层折叠中完整训练模型的一个循环, 而不重新初始化参数, 使模型能够通过从不同数据子集学习逐步优化其参数。具体来说, 模型首先在内层折叠 1 上进行训练, 然后继续在

在超参数优化的第一部分中, 外部交叉验证循环 (图 1a)。以下方程定义了我们从 PyTorch geometric (版本 2.5.3) 实现的卷积层。

GCN 使用归一化邻接矩阵从节点的邻居聚合信息。在公式1中, $\mathbf{h}_i^{(l)}$ 是第 l 层节点 i 的嵌入, $\mathcal{N}(i)$ 表示节点 i 的邻居集合, $\hat{d}_i = 1 + \sum_{j \in \mathcal{N}(i)} e_{j,i}$ 是包括自环在内的节点 i 的度, 其中 $e_{j,i}$ 表示节点 j 和 i 的边权重。 $\mathbf{W}^{(l)}$ 是第 l 层的可训练权重矩阵, σ 是激活函数:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{\hat{d}_i \hat{d}_j}} \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right) \quad (1)$$

GAT 引入注意力机制来权衡邻居节点的贡献。在下面的公式 2 中, $\alpha_{ij}^{(l)}$ 是节点 i 和 j 之间的注意力系数, 取代了公式 1 中的度归一化:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{ij}^{(l)} \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right) \quad (2)$$

方程2中的注意力系数使用下方方程3中的 LeakyReLU 激活函数进行计算 (并归一化), 其中 $\mathbf{a}^{(l)}$ 是可学习的注意力向量, $\parallel$ 表示连接:

$$\alpha_{ij}^{(l)} = \frac{\exp \left(\text{LeakyReLU} \left(\mathbf{a}^{(l)T} \left[\mathbf{W}^{(l)} \mathbf{h}_i^{(l)} \parallel \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right] \right) \right)}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp \left(\text{LeakyReLU} \left(\mathbf{a}^{(l)T} \left[\mathbf{W}^{(l)} \mathbf{h}_i^{(l)} \parallel \mathbf{W}^{(l)} \mathbf{h}_k^{(l)} \right] \right) \right)} \quad (3)$$

GraphSAGE 通过使用均值聚合函数从邻居节点聚合信息来更新节点嵌入。在方程 4 中, $\mathbf{W}_1^{(l)}$ 和 $\mathbf{W}_2^{(l)}$ 是第 l 层的可训练权重矩阵, 均值聚合确保邻居节点的贡献被节点 i 的邻居数量归一化:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\mathbf{W}_1^{(l)} \mathbf{h}_i^{(l)} + \mathbf{W}_2^{(l)} \cdot \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right) \quad (4)$$

最后, 在更新权重时, GIN 使用求和聚合, 然后通过一个可训练的多层感知器 (MLP) 来提高节点表示的表达能力。在公式5中, ϵ 是一个固定的标量参数, MLP 表示应用于聚合特征的多层感知器:

$$\mathbf{h}_i^{(l+1)} = \text{MLP} \left((1 + \epsilon) \cdot \mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right) \quad (5)$$

模型架构

实验设置包括实现和评估四种图神经网络 (GNN) 架构: 图卷积网络 (GCN)、图注意力网络 (GAT)、GraphSAGE 和图同构网络 (GIN)。每种架构都通过在内部交叉验证循环中进行超参数网格搜索来优化数据集, 并在未见过的测试数据集上进行测试。

Model reinitialisation and sequential warm starts for fine tuning

After identifying the optimal hyperparameters, we addressed the variability introduced by random weight initialisation by training models with 20 different random seeds from PyTorch (version 2.3.1) [59]. For each outer fold, we selected the best-performing seed based on the highest combined rank of median and minimum validation AUC scores obtained from the inner folds. This model reinitialisation step aimed to enhance performance stability across all inner folds and ensured the reproducibility for the further work. To further refine the models, we employed a sequential warm starts method for fine-tuning the parameters (see Fig. 1b). Each run in the sequential warm starts method consisted of a complete cycle of training the model across all inner folds without reinitialising the parameters, allowing the model to incrementally refine its parameters by learning from different subsets of the data. Specifically, the model is first trained on inner fold 1. It then continues training on inner fold 2 using the learned parameters from fold 1, and this process repeats through folds 3, 4, and 5. Each run yielded five validation AUC scores per run per outer fold, resulting in a total of 20 validation AUC scores per run across all four outer folds. We repeated the sequential warm starts process for at least three runs to assess whether continued training led to significant improvements. Upon the completion of each run, we conducted significance testing using the Wilcoxon rank sum test to compare the current run's AUC scores with those from the previous two runs [60]. We applied a significance level of $\alpha = 0.05$, which was then adjusted using the Sidak correction for two comparisons to control the family-wise error rate [61]. If no statistically significant improvement was observed (i.e. both p-values exceeded the corrected α), we halted the sequential warm starts for that model. Throughout the training process, we applied an early stopping function that monitored the validation AUC scores and halted training for a fold if no improvement was observed over 50 epochs. The statistical significance testing and the early stopping prevented further training beyond the point of meaningful improvement and model overfitting to a certain inner fold validation set to enhance generalisability of the GNN models.

Model architecture

The experimental setup involved implementing and evaluating four Graph Neural Network (GNN) architectures: Graph Convolutional Network (GCN), Graph Attention Network (GAT), GraphSAGE, and Graph Isomorphism Network (GIN). Each architecture was optimised for the dataset using a hyperparameter grid search within the inner CV loop and tested on the unseen test dataset in

the outer CV loop within the first part of the hyperparameter optimisation (Fig. 1a). The following equations define the convolutional layers we implemented from PyTorch geometric (version 2.5.3).

GCN aggregates information from a node's neighbours using normalised adjacency matrix. In equation 1, $\mathbf{h}_i^{(l)}$ is the embedding of node i at layer l , $\mathcal{N}(i)$ denotes the set of neighbours of node i , and $\hat{d}_i = 1 + \sum_{j \in \mathcal{N}(i)} e_{j,i}$ is the degree of node i including self-loops, where $e_{j,i}$ signifies the edge weight of nodes j and i . $\mathbf{W}^{(l)}$ is the trainable weight matrix for layer l , and σ is the activation function:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{\hat{d}_i \hat{d}_j}} \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right) \quad (1)$$

GAT introduces attention mechanisms to weight the contributions of neighbouring nodes. In equation 2 below, $\alpha_{ij}^{(l)}$ is the attention coefficient between nodes i and j that replaces the degree normalisation in equation 1:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{ij}^{(l)} \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right) \quad (2)$$

The attention coefficient from equation 2 is computed (and normalised) using a LeakyReLU activation function in equation 3 below, where $\mathbf{a}^{(l)}$ is a learnable attention vector and $\parallel$ denotes concatenation:

$$\alpha_{ij}^{(l)} = \frac{\exp \left(\text{LeakyReLU} \left(\mathbf{a}^{(l)T} \left[\mathbf{W}^{(l)} \mathbf{h}_i^{(l)} \parallel \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right] \right) \right)}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp \left(\text{LeakyReLU} \left(\mathbf{a}^{(l)T} \left[\mathbf{W}^{(l)} \mathbf{h}_i^{(l)} \parallel \mathbf{W}^{(l)} \mathbf{h}_k^{(l)} \right] \right) \right)} \quad (3)$$

GraphSAGE updates node embeddings by aggregating information from neighbouring nodes using a mean aggregation function. In equation 4, $\mathbf{W}_1^{(l)}$ and $\mathbf{W}_2^{(l)}$ are trainable weight matrices at layer l , and the mean aggregation ensures that the contributions from neighbours are normalised by the number of neighbours of node i :

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\mathbf{W}_1^{(l)} \mathbf{h}_i^{(l)} + \mathbf{W}_2^{(l)} \cdot \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right) \quad (4)$$

Finally, for updating weights, GIN employs sum aggregation followed by a trainable multilayer perceptron (MLP) to improve expressive capacity of node representations. In equation 5, ϵ is a fixed scalar parameter, and MLP represents an MLP applied to the aggregated features:

$$\mathbf{h}_i^{(l+1)} = \text{MLP} \left((1 + \epsilon) \cdot \mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right) \quad (5)$$

广义 **GNN** 模型
通用的 GNN 模型类采用模块化架构, 允许指定不同的卷积层类型。我们实现了以下卷积类型: GCNConv、GATConv 和 SAGEConv。该模型包含三层卷积层, 每层后都跟有一个 ReLU 激活函数。架构如下所示:

• 权重初始化: 对所有网络权重应用了Xavier初始化以增强收敛性 [62]。 • 卷积层:

– 第1层: 卷积层, 输出维度为 hidden_通道*2, 随后进行 ReLU 激活。 – 第2层: 卷积层, 输出维度为 hidden_通道, 随后进行 ReLU 激活。 – 第3层: 卷积层, 输出维度为 hidden_通道/2, 随后进行 ReLU 激活。

• Dropout 层: 在最后一个卷积层之后使用了 dropout 率以减轻过拟合。 • 全局池化: 应用全局平均池化将节点特征聚合成固定大小的图级表示。 • 输出层: 使用线性层将池化后的表示映射到单个输出 logit。在使用 BCEWithLogitsLoss 或 CrossEntropyLoss 进行 ClinTox 分类任务的损失计算过程中, 加入了 sigmoid 激活函数。

图同构网络 (**GIN**) 模型
GIN 模型采用了专门的架构, 利用 GINConv 层。每个 GINConv 层都使用多层感知机 (MLP) 进行特征变换。模型架构总结如下:

• 权重初始化: 对所有权重应用了Xavier初始化, 以促进训练的稳定性。 • GIN卷积层:

– 层 1: 一个 GINConv 层, 其 MLP 包含两个线性层和 ReLU 激活, 将特征转换为 hidden_ 通道*2。

– 第2层: 一个GINConv层, 带有一个将特征转换为 hidden_通道的MLP。 – 第3层: 一个GINConv层, 带有一个将特征尺寸减少到hidden_通道/2的MLP。

• Dropout 层: 在最后的 GIN 卷积层之后采用了 dropout 率, 以减少过拟合。 • 全局池化: 应用全局平均池化以获得固定大小的图级表示。 • 输出层: 线性层将池化后的表示映射到最终的输出 logit, sigmoid 激活函数由 ClinTox 分类任务中的 BCEWithLogitsLoss 或 CrossEntropyLoss 处理。

评估指标
在顺序热启动之后, 模型使用多种指标在每个外层折的测试集上进行了评估。准确率衡量了正确分类实例的比例。F1分数作为精确度和召回率的调和平均值, 用于评估精确度与敏感性之间的平衡。ROC曲线下面积 (AUC) 评估了模型区分类别的能力, 而马修相关系数 (MCC) 通过考虑真阳性、假阳性、真阴性和假阴性提供了平衡的评估。虽然中位数、最大值和最小值可能更适合用于较少的评估指标以更好地捕捉变异性, 但我们选择了均值和标准差值, 因为其他最新模型报告了这些值 [4, 17, 18]。为了便于将我们的结果与先前以及未来在相同数据集上的工作进行比较, 我们报告了测试集 (作为第一步从剩余数据中划分, 见图1) 的所有指标结果。

结果
处理过的美国 FDA DILI 数据集及与以往研究的比较
在最新发布的 FDA DILI 数据集 [11] 中有 1,279 种药物。我们无法获取 SMILES

Generalised GNN model
The generalised GNN model class utilised a modular architecture that allows for the specification of different convolutional layer types. We implemented the following convolutional types: GCNConv, GATConv, and SAGEConv. The model comprised three convolutional layers, each followed by a ReLU activation function. The architecture is detailed below:

- **Weight Initialisation:** Xavier initialisation was applied to all network weights to enhance convergence [62].
- **Convolutional Layers:**

- **Layer 1:** Convolutional layer with output dimensions of hidden_channels*2, followed by ReLU activation.
- **Layer 2:** Convolutional layer with output dimensions of hidden_channels, followed by ReLU activation.
- **Layer 3:** Convolutional layer with output dimensions of hidden_channels/2, followed by ReLU activation.

- **Dropout Layer:** A dropout rate was employed after the final convolutional layer to mitigate overfitting.
- **Global Pooling:** Global mean pooling was applied to aggregate node features into a fixed-size graph-level representation.
- **Output Layer:** A linear layer was used to map the pooled representation to a single output logit. The sigmoid activation function was incorporated during the loss computation using BCEWithLogitsLoss or CrossEntropyLoss for the ClinTox classification task.

Graph isomorphism network (GIN) model
The GIN model featured a specialised architecture leveraging GINConv layers. Each GINConv layer employed a multi-layer perceptron (MLP) for feature transformation. The model architecture is summarised as follows:

- **Weight Initialisation:** Xavier initialisation was applied to all weights to promote stable training.
- **GIN Convolutional Layers:**

- **Layer 1:** A GINConv layer with an MLP comprising two linear layers and ReLU activation, transforming features to hidden_channels*2.

- **Layer 2:** A GINConv layer with an MLP that transforms features to hidden_channels.
- **Layer 3:** A GINConv layer with an MLP reducing the feature size to hidden_channels/2.

- **Dropout Layer:** A dropout rate was employed after the final GIN convolutional layer to reduce overfitting.
- **Global Pooling:** Global mean pooling was applied to derive a fixed-size graph-level representation.
- **Output Layer:** A linear layer mapped the pooled representation to the final output logit, with the sigmoid activation function handled by the BCEWithLogitsLoss or CrossEntropyLoss for the ClinTox classification task.

Evaluation metrics
Following the sequential warm starts, the models were evaluated on the test set of each outer fold using several metrics. Accuracy measured the proportion of correctly classified instances. The F1-score, calculated as the harmonic mean of precision and recall, assessed the balance between precision and sensitivity. The Area-Under-the-ROC-Curve (AUC) evaluated the model's ability to distinguish between classes, while Matthew's Correlation Coefficient (MCC) provided a balanced assessment by accounting for true and false positives and negatives. While median, maximum, and minimum values might be more suitable for a lower number of evaluation metrics to better capture variability, we chose mean and standard deviation values, as other state-of-the-art models reported these values [4, 17, 18]. To allow comparison of our results to prior as well as future work on the same datasets, we report results on all metrics for the test set (split from the remainder of the data as first step, Fig. 1). AUC is commonly used by others to show the performance of binary classifiers, however it is not suitable for imbalanced datasets (whereas F1-score and MCC are more suitable for reporting on datasets with class imbalances). We report the AUC to allow comparison of our results to work only reporting AUC. The Area Under the Precision-Recall Curve (AUC-PR) is not reported here as this is more suited for datasets where the positive class is rare opposed to the imbalance seen in the datasets we demonstrate the method on.

Results
Processed US FDA DILI dataset and comparison with previous studies
There are 1,279 drugs in the latest release of the FDA DILI dataset [11]. We were unable to retrieve SMILES

表2 不同DILI研究中药物数量及阳性对阴性 (+ 比-) 比率 的比较

研究	DILI 药物	+ 与- 的比例
美国 FDA DIList [11]	一千二百七十九	1.50:1
我们的数据	一千一百六十七	1.63:1
DILIPredictor [18]	一千一百一十一	1.81:1
DeepDILI [4]; SSM [17]	一千二	1.52:1

该表突出了不同 DILI 研究中数据集大小和类别平衡的差异。这些研究包括原始美国 FDA DILI 数据集、我们处理过的数据集、DILIPredictor、DeepDILI 以及 SSM。我们的数据集保留了原始 FDA 数据集中的更多药物，其阳性与阴性比例更接近原始数据，表明与 DILIPredictor 研究相比，数据集更加平衡。

使用请求方法获取33种DILI阳性药物和45种DILI阴性药物的字符串。无法检索这些字符串的主要原因是这些药物要么是大分子药物（例如，奥米妥珠单抗、促胰液素），要么涉及金属置换疗法（例如，酚磺乙胺），或者由于具有不同的结构变体。然后，对1,201种药物进行了预处理和分子标准化。在这些步骤之后，剩下 1,167种药物，并且在图生成步骤中，这些药物全部通过了分子优化过程。

我们的数据处理策略相比之前使用 FDA 发布的 DILI 数据集的研究，从 FDA DILI 数据集中得到了更多药物（表 2）。最新的 FDA DILI 数据集称为 DIList [11], , 包含 1,279 种药物，正负比例为 1.50:1。DeepDILI 在排除了生物制剂、药物混合物、有机金属化合物和无机化合物后使用了 1,002 种药物 [4], , 而 SSM 使用了来自 DeepDILI 研究的相同数据集 [17]。DILIPredictor 通过结合最新版本（DIList [11]）和之前版本（DILIRank [10]）的 FDA DILI 数据集进行了改进。在应用其分子标准化方法后，DILIPredictor 数据集包含 1,111 种药物，正负比例为 1.81:1，相比原始数据集更不平衡 [18]。相比之下，我们的预处理和分子标准化策略产生了 1,167 种药物，利用了 FDA DILI 数据集中的更多信息，实现了正负比例为 1.63:1。

此外，为了评估我们的分子标准化方法在分子优化后对分子键长变化的影响，我们计算了标准化和非标准化分子键长变化的统计指标（平均值、中位数、标准差），如补充表2所示。我们观察到，分子标准化方法将最大键长变化的极端异常值从14.44%降低到7.45%，标准差从0.62%降至0.53%。尽管标准化和非标准化分子的平均键长变化差异在统计上不显著（p=0.265），应用分子标准化方法对确保计算化学中的数据质量和可重复性是有益的 [63]。用于此分析的数据，包括药物分子优化后的键长，可在我们的GitHub上获取。

模型在DILI预测任务中的表现

表3总结了四种GNN模型在DILI预测任务中，在我们嵌套交叉验证（CV）框架的不同阶段的表现，以及它们的测试性能指标。在超参数优化之后，所有模型显示出相当的初始性能，平均验证AUC得分在0.624到0.627之间。在模型重新初始化期间，所有模型的性能都有所提高，平均验证AUC的提升范围为0.043到0.068。GAT和GraphSAGE的性能提升最大。

通过顺序热启动方法的进一步优化产生了最高的性能提升。GraphSAGE 达到最佳验证性能，平均 AUC 为 0.953，其次是 GIN（0.916）和 GCN（0.881）。GAT 显示出适度的提升，达到 0.728。性能提升的差异很可能受到每个模型重复运行次数不同的影响（GCN：4，GAT：1，GraphSAGE：3，GIN：5，见补充图 1）。具体而言，由于统计显著性检验表明没有进一步显著改进，GAT 的顺序热启动比其他模型提前停止。

在测试集上，GraphSAGE 在所有评估指标上都取得了最高的性能，包括测试 AUC（0.897 ± 0.043）、准确率（0.836 ± 0.095）、F1 分数（0.846 ± 0.106）和 MCC（0.698 ± 0.148），其次是 GIN 模型。各模型在测试性能上的差异与在顺序预热启动方法中获得的验证 AUC 分数的趋势一致。这些结果表明，顺序预热启动方法可以有效地提升模型性能，通过

Table 2 Comparison of the number of drugs and the positive-to-negative (+ to -) ratios in different DILI studies

Studies	DILI drugs	+ to - ratio
US FDA DIList [11]	1,279	1.50:1
Our data	1,167	1.63:1
DILIPredictor [18]	1,111	1.81:1
DeepDILI [4]; SSM [17]	1,002	1.52:1

The table highlights the differences in dataset sizes and class balance across different DILI studies. The studies include the original US FDA DILI dataset, our processed dataset, DILIPredictor, DeepDILI, and SSM. Our dataset retains a greater number of drugs from the original FDA dataset, with a positive-to-negative ratio closer to the original, indicating a more balanced dataset compared to the DILIPredictor study

strings for 33 DILI positive and 45 DILI negative drugs using the request method. The failure to retrieve these strings is primarily due to the drugs being either large biomolecule drugs (e.g., Gemtuzumab ozogamicin, Secretin) or involved in metal replacement therapy (e.g., Ferumoxytol), or due to having different structural variations. Then 1,201 drugs went through preprocessing and molecular standardisation. Following these steps, 1,167 drugs remained and all of these passed the molecular optimisation process during the graph generation step.

Our data processing strategies have yielded more drugs from the FDA DILI dataset compared to previous studies that used FDA-released DILI datasets (Table 2). The latest FDA DILI dataset is called DIList [11], which contains 1,279 drugs with a positive to negative ratio of 1.50:1. DeepDILI used 1,002 drugs after excluding biologics, drug mixtures, organometallics, and inorganics [4], and SSM used the same dataset from the DeepDILI study [17]. DILIPredictor improved upon this by combining the latest (DIList [11]) and the previous version (DILIRank [10]) of the FDA DILI dataset. After applying their molecular standardisation method, DILIPredictor dataset includes 1,111 drugs with a more imbalanced positive to negative ratio of 1.81:1 compared to the original dataset [18]. In contrast, our pre-processing and molecular standardisation strategies resulted in 1,167 drugs, leveraging more information from the FDA DILI dataset and achieving a positive to negative ratio of 1.63:1. This indicates that our dataset is closer to the balance seen in the original dataset compared to the DILIPredictor dataset. We identified that 8 drugs from the DeepDILI study and 122 drugs (including 37 drugs from DILIRank) from the DILIPredictor study are not present in our dataset. These drugs, along with those for which SMILES retrieval failed, are available in our GitHub. The DNN [19] and DNN-GATNN [20] studies combined LiverTox dataset with the FDA DILI dataset; therefore, they were not included in this study.

Furthermore, to assess the effects of our molecular standardisation method on molecular bond length changes following molecular optimisation, we computed statistical metrics (mean, median, standard deviation) of bond length changes for both standardised and non-standardised molecules, as shown in Supplementary Table 2. We observed that our molecular standardisation method reduces extreme outliers in maximum bond length changes from 14.44% to 7.45% and standard deviation decreasing from 0.62% to 0.53%. Although the difference in mean bond length change between standardised and non-standardised molecules is not statistically significant (p=0.265), applying the molecular standardisation method is beneficial for ensuring data quality and reproducibility in computational chemistry [63]. The data used for this analysis, including bond lengths after optimisation of drug molecules, is available in our GitHub.

Model performance in the DILI prediction task

Table 3 summarises the performance of the four GNN models on the DILI prediction task across different stages of our Nested Cross-Validation (CV) framework, along with their test performance metrics. After hyperparameter optimisation, all models demonstrated comparable initial performance with mean validation AUC scores ranging between 0.624 and 0.627. During model reinitialisation, performance improved for all models, with increases in mean validation AUC ranging from 0.043 to 0.068. The largest improvements were observed for GAT and GraphSAGE.

Further refinement through the sequential warm starts method produced the highest performance improvements. GraphSAGE achieved the best validation performance, with a mean AUC of 0.953, followed by GIN (0.916) and GCN (0.881). GAT showed a modest improvement, reaching 0.728. This discrepancy in improvement is likely influenced by the different numbers of repeated runs performed for each model (GCN: 4, GAT: 1, GraphSAGE: 3, and GIN: 5, see Supplementary Figure 1). Specifically, GAT’s sequential warm starts halted earlier than the other models since statistical significance testing concluded that there was no further significant improvement.

On the test sets, GraphSAGE achieved the highest performance across all evaluation metrics, including test AUC (0.897 ± 0.043), accuracy (0.836 ± 0.095), F1-score (0.846 ± 0.106), and MCC (0.698 ± 0.148), followed by the GIN model. The variation in test performance among the models aligns with the trends observed in their validation AUC scores obtained during the sequential warm starts method. These results indicate that the sequential warm starts method effectively enhances model performance by

表 3 GNN 模型在 DILI 预测任务中的性能指标 (均值 ± 标准差)

公制	GCN	GAT	GIN	图SAGE
验证AUC				
超参数优化	0. 627 ± 0.034	0. 627 ± 0.031	0.626 ± 0.032	0.624 ± 0.032
模型重新初始化	0.670 ± 0.037	0. 695 ± 0.047	0.671 ± 0.033	0.692 ± 0.043
顺序热启动	0.881 ± 0.062	0.728 ± 0.054	0.916 ± 0.051	0.953 ± 0.061
测试AUC	0.847 ± 0.058	0.728 ± 0.059	0.874 ± 0.027	0.897 ± 0.043
测试准确性	0.733 ± 0.081	0.619 ± 0.049	0.798 ± 0.058	0.836 ± 0.095
测试F1分数	0.735 ± 0.098	0.611 ± 0.069	0.814 ± 0.062	0.846 ± 0.106
测试 MCC	0.533 ± 0.124	0.323 ± 0.078	0.628 ± 0.094	0.698 ± 0.148

该表突出了在嵌套交叉验证 (CV) 框架不同阶段的验证 AUC 分数, 以及 GCN、GAT、GIN 和 GraphSAGE 模型的测试性能指标。验证 AUC 报告了用于超参数优化、模型重新初始化以及应用顺序热启动方法之后的结果。这些步骤是相互累积的, 模型重新初始化中使用了最优超参数, 随后顺序热启动使用最佳模型, 以逐步提升模型性能。测试指标包括 AUC、准确率、F1 分数和 MCC, 以平均值 ± 标准差的形式报告。我们的方法通过在训练子集上进行渐进式学习并使用统计显著性测试来优化泛化能力, 从而提升了性能。每个指标中的最佳平均值以粗体突出显示。

表 4 使用基本模型与 GNN 模型的测试 AUC 分数比较 (均值 ± 标准差) . 跨四个数据集的自定义功能

数据集	功能集	GCN	GAT	GIN	图SAGE
DILI	基础	0.537 ± 0.022	0.551 ± 0.022	0.511 ± 0.028	0.723 ± 0.045
	自定义	0.847 ± 0.058	0.728 ± 0.059	0.874 ± 0.027	0.897 ± 0.043
临床毒理	基础	0.502 ± 0.072	0.486 ± 0.071	0.462 ± 0.054	0.723 ± 0.045
	自定义	0.918 ± 0.028	0.901 ± 0.043	0.897 ± 0.055	0.871 ± 0.051
BBBP	基础	0.500 ± 0.101	0.358 ± 0.028	0.517 ± 0.074	0.871 ± 0.019
	自定义	0.990 ± 0.006	0.973 ± 0.020	0.981 ± 0.012	0.993 ± 0.006
BACE	基础	0.515 ± 0.061	0.541 ± 0.083	0.518 ± 0.040	0.604 ± 0.031
	自定义	0.906 ± 0.053	0.845 ± 0.047	0.620 ± 0.066	0.953 ± 0.020

该表显示了每种GNN在四个不同数据集上的测试AUC分数: DILI、ClinTox、BBBP和BACE。对于每个模型和数据集, 结果分别展示了基础分子图特征 (原子序数和键类型) 和我们自定义设计的特征, 包括丰富的化学和结构信息 (见表1)。每个数据集的最高AUC分数用加粗表示

利用对训练子集的渐进式学习和统计显著性测试来优化泛化。

其他预测任务中的模型表现: 基本特征与我们的自定义特征

表4比较了我们GNN模型在使用基本化学图与我们定制分子图时, 在DILI、ClinTox、BBBP和BACE四个数据集上的测试AUC得分。该分析评估了化学特征工程对不同预测任务模型性能的有效性, 包括毒性 (DILI 和ClinTox)、渗透性 (BBBP) 和生物化学活性 (BACE)。

在使用基本特征集 (仅使用原子序数和键类型作为节点和边特征) 的情况下, 模型通常表现出接近 0.5 的低 AUC 分数, 表明预测性能有限。具体而言, GCN、GAT 和 GIN 的性能最小, AUC 范围约为 0.358 到 0.551。

这些数据集。其中一个例外是 GraphSAGE, 它在 DILI 数据集上取得了相对较高的 AUC 分数 0.723, 在 ClinTox 上为 0.739, 在 BBBP 上为 0.871。这种性能的提高可以归因于 GraphSAGE 的归纳学习方法, 它能够有效地聚合邻近节点的信息, 以捕捉图中的结构模式和关系, 部分弥补基本分子图中有限特征的不足。然而, 在 BACE 数据集上, 即使是 GraphSAGE 的表现也较为有限, AUC 为 0.604, 这表明在这一生化活性预测任务中, 基本特征不足以区分活性化合物。

相比之下, 使用定制设计的特征 (包括更丰富的化学和结构信息) 在所有数据集上都显著提高了 AUC 值。对于 DILI 数据集, GIN 的提升最高, AUC 从使用基础特征的 0.511 提升到使用定制特征的 0.874, 反映出 0.363 的提升。对于 Clintox

Table 3 Performance Metrics (mean ± std) of GNN Models on the DILI Prediction Task

Metric	GCN	GAT	GIN	GraphSAGE
Validation AUC				
Hyperparameter optimisation	0.627 ± 0.034	0.627 ± 0.031	0.626 ± 0.032	0.624 ± 0.032
Model reinitialisation	0.670 ± 0.037	0.695 ± 0.047	0.671 ± 0.033	0.692 ± 0.043
Sequential warm starts	0.881 ± 0.062	0.728 ± 0.054	0.916 ± 0.051	0.953 ± 0.061
Test AUC	0.847 ± 0.058	0.728 ± 0.059	0.874 ± 0.027	0.897 ± 0.043
Test accuracy	0.733 ± 0.081	0.619 ± 0.049	0.798 ± 0.058	0.836 ± 0.095
Test F1-score	0.735 ± 0.098	0.611 ± 0.069	0.814 ± 0.062	0.846 ± 0.106
Test MCC	0.533 ± 0.124	0.323 ± 0.078	0.628 ± 0.094	0.698 ± 0.148

The table highlights validation AUC scores at different stages of the Nested Cross-Validation (CV) framework and test performance metrics for GCN, GAT, GIN, and GraphSAGE models. Validation AUCs are reported for hyperparameter optimisation, model reinitialisation, and after applying the Sequential Warm Starts method. These steps build upon each other, with the optimal hyperparameters used in the model reinitialisation and the best models subsequently used for the sequential warm starts to progressively enhance the model performance. The test metrics include AUC, accuracy, F1-score, and MCC, reported as mean ± standard deviation. Our method improved performance by leveraging progressive learning on training subsets and statistical significance testing to optimise generalisation. The best mean values in each metric are highlighted in bold

Table 4 Comparison of test AUC scores (mean ± std) for GNN models using basic vs. custom features across four datasets

Dataset	Feature Set	GCN	GAT	GIN	GraphSAGE
DILI	Basic	0.537 ± 0.022	0.551 ± 0.022	0.511 ± 0.028	0.723 ± 0.045
	Custom	0.847 ± 0.058	0.728 ± 0.059	0.874 ± 0.027	0.897 ± 0.043
ClinTox	Basic	0.502 ± 0.072	0.486 ± 0.071	0.462 ± 0.054	0.723 ± 0.045
	Custom	0.918 ± 0.028	0.901 ± 0.043	0.897 ± 0.055	0.871 ± 0.051
BBBP	Basic	0.500 ± 0.101	0.358 ± 0.028	0.517 ± 0.074	0.871 ± 0.019
	Custom	0.990 ± 0.006	0.973 ± 0.020	0.981 ± 0.012	0.993 ± 0.006
BACE	Basic	0.515 ± 0.061	0.541 ± 0.083	0.518 ± 0.040	0.604 ± 0.031
	Custom	0.906 ± 0.053	0.845 ± 0.047	0.620 ± 0.066	0.953 ± 0.020

This table presents the test AUC scores for each GNN on four different datasets: DILI, ClinTox, BBBP, and BACE. For each model and dataset, results are shown separately for basic molecular graph features (atomic number and bond type) and our custom-designed features, which include an enriched set of chemical and structural information (see Table 1). The highest AUC scores for each dataset are highlighted in bold

leveraging progressive learning on training subsets and statistical significance testing to optimise generalisation.

Model performance in other prediction tasks: basic features vs our custom features

Table 4 compares the test AUC scores of our GNN models using basic chemical graphs versus our custom-designed molecular graphs across four datasets: DILI, ClinTox, BBBP, and BACE. This analysis evaluates the effectiveness of chemical feature engineering on model performance in different prediction tasks, including toxicity (DILI and ClinTox), permeability (BBBP), and biochemical activity (BACE).

In the basic feature set, which utilises only atomic number and bond type as node and edge features, the models generally exhibited low AUC scores close to 0.5, indicating limited predictive performance. Specifically, GCN, GAT, and GIN showed minimal performance, with AUCs ranging from approximately 0.358 to 0.551 across

the datasets. An exception was observed with GraphSAGE, which achieved comparatively higher AUC scores of 0.723 on the DILI dataset, 0.739 on ClinTox, and 0.871 on BBBP. This improved performance can be attributed to GraphSAGE's inductive learning approach, which effectively aggregates information from neighbouring nodes to capture structural patterns and relationships within the graphs, partially compensating for the limited features in the basic molecular graphs. However, on the BACE dataset, even GraphSAGE's performance was modest, with an AUC of 0.604, suggesting that the basic features were insufficient for distinguishing active compounds in this biochemical activity prediction task.

In contrast, the use of the custom-designed features, which include a richer set of chemical and structural information, resulted in significant improvements in AUC values across all datasets. For the DILI dataset, GIN showed the highest improvement, with an AUC increase from 0.511 using basic features to 0.874 with custom features, reflecting an improvement of 0.363. For the Clintox

在数据集中, GCN 显示了最显著的提升, AUC 从使用基本特征的 0.502 提升到使用自定义特征的 0.918, 增加了 0.416。对于 BBBP 数据集, GCN 在 AUC 值上表现出最大的提升, 从使用基本特征的 0.500 提升到使用自定义特征的 0.990, 标志着显著的 0.490 增长。对于 BACE 数据集, GraphSAGE 实现了最高的提升, AUC 值从基本特征集的 0.604 上升到自定义特征的 0.953, 提升了 0.349。

总体而言, GraphSAGE 在 DILI、BBBP 和 BACE 预测任务中表现最佳, 而 GCN 在 ClinTox 预测任务中取得了最高的性能。这些表现最好的模型将在讨论部分与最先进的图分类模型进行比较。

讨论

嵌套交叉验证框架对模型性能的影响 本研究中采用的嵌套交叉验证 (CV) 框架通过系统地选择最优超参数组合、减少随机权重初始化带来的变异性, 并在内层 CV 循环中对不同数据子集进行顺序热启动来微调模型参数, 从而显著提高了我们 GNN 模型的性能。在超参数优化阶段, 我们通过使用中位数和最小验证 AUC 分数评估所有五个内折来选择最优超参数组合。这种方法确保所选择的超参数组合不会过度针对特定内折进行调优, 而是在不同的数据划分中表现良好, 从而最大限度地减少选择次优超参数组合的可能性 (见补充图2)。在我们对模型重新初始化方法的初步研究中, 我们使用5个不同的随机种子进行了10折交叉验证, 并观察到

结果显示, 与超参数优化阶段相比, 在 DILI 预测中, 四种 GNN 模型的验证 AUC 评分均有所提高, 增幅在 0.043 到 0.068 之间 (见表3)。顺序热启动方法通过在训练数据的不同子集 (内部折分) 上逐步训练模型, 进一步提升了模型性能。该方法相比传统训练方法具有优势, 因为它解决了早停函数引入的限制。当只使用单一验证集时, 模型可能会过拟合验证集中的特定模式, 或在性能上出现暂时的平台期, 导致过早终止, 从而阻碍模型充分捕捉训练数据中的 DILI 特征。顺序热启动通过在不同的内部折分上逐步优化模型参数解决了这一问题, 确保了更稳健的训练和更好的泛化性能。此外, Wilcoxon 秩和检验

我们定制设计的分子图的有效性

我们的实验结果表明, 使用定制设计的分子图显著提升了 GNN 模型在各种预测任务中的性能。如表4所示, 定制特征集在所有模型和数据集上均比基础特征集产生了更高的 AUC 值。这种持续的提升强调了在不同化学空间中捕捉复杂分子特征时, 纳入先进的化学和结构特征的重要性, 包括毒性 (DILI 和 ClinTox)、渗透性 (BBBP) 和生化活性 (BACE)。通过包含来自 3D 优化分子表示的详细化学和结构属性, 丰富的特征集为 GNN 模型提供了更具信息量和判别力的分子图表示。因此, 模型能够有效地学习

dataset, GCN displayed the most notable improvement, with the AUC improving from 0.502 with basic features to 0.918 with custom features, an increase of 0.416. For the BBBP dataset, GCN showed the greatest improvement in AUC values, improving from 0.500 with basic features to 0.990 with custom features, marking a substantial increase of 0.490. For the BACE dataset, GraphSAGE achieved the highest improvement, with AUC values rising from 0.604 in the basic feature set to 0.953 with the custom features, with an improvement of 0.349.

Overall, GraphSAGE was the best-performing model in the DILI, BBBP, and BACE prediction tasks, while GCN achieved the highest performance in the ClinTox prediction task. These best-performing models will be compared to the state-of-the-art graph classification models in the discussion section.

Discussion

Impact of the nested CV framework on model performance

The nested CV framework employed in this study significantly improved the performance of our GNN models by systematically selecting optimal hyperparameter combinations, reducing variability from random weight initialisation, and fine-tuning the model parameters with sequential warm starts on different subsets of the data within the inner CV loop. In the hyperparameter optimisation stage, we selected the optimal hyperparameter combination by evaluating on all five inner folds using both median and minimum validation AUC scores. This approach ensured that the selected hyperparameter combination was not overly tuned to a specific inner fold but generalised well across diverse data splits therefore minimising the likelihood of selecting sub-optimal hyperparameter combinations (see Supplementary Figure 2). In our preliminary investigation of the model reinitialization method, we ran 10-fold CV with 5 different seeds and observed performance discrepancies due to random weight initialization (see Supplementary Figure 3). This outcome is particularly relevant in deep learning models that are sensitive to local minima where variations in initial weights and biases can lead to differences in performance outcome. To address this variability, we trained the models with 20 different seeds in the main study. For each outer fold, the best-performing seed was selected based on a combined ranking of the median and minimum validation AUC scores across all five inner folds (see Supplementary Figure 4). This approach further ensured stability of the model performance by reducing bias towards certain inner folds where the early stopping function might have halted training prematurely or prolonged it due to specific data characteristics. As a

result, we observed increases in validation AUC scores between 0.043 and 0.068 across all four GNN models in DILI prediction in comparison to the hyperparameter optimisation stage (see Table 3). The sequential warm starts method further improved the model performance by progressively training the models on different subsets; inner fold splits, of the training data. This method provided advantages over conventional training approaches by addressing limitations introduced by early stopping functions. When a single validation set is used, models may overfit to specific patterns in the validation set or face temporary plateaus in performance, leading to premature termination which prevent the model from fully capturing the DILI features from the training data. Sequential warm starts addressed this issue by incrementally refining model parameters across the different inner fold splits, ensuring more robust training and improved generalisation. In addition, Wilcoxon Rank Sum Test ensured that the repeated runs in sequential warm starts continued only when significant improvements were observed from the two previous runs. This prevented unnecessary training cycles and prevented overfitting. We observed that there is an increase of 0.132 in test AUC for GraphSAGE model when employing the sequential warm starts method over the conventional methods (see Supplementary Table 3). Overall, our method systematically improved the generalisability and stability of the GNN models during training. Hyperparameter optimisation and random seed reinitialisation laid a solid foundation for model fine-tuning with the sequential warm starts which yielded the best performance metrics across all four GNN models.

Effectiveness of our custom-designed molecular graphs

Our experimental results demonstrate that the use of custom-designed molecular graphs significantly enhances the performance of GNN models across various prediction tasks. As shown in Table 4, the custom feature sets consistently yielded higher AUC values compared to the basic feature sets across all models and datasets. This consistent improvement underscores the importance of incorporating advanced chemical and structural features in capturing complex molecular characteristics across different chemical spaces, including toxicity (DILI and ClinTox), permeability (BBBP), and biochemical activity (BACE). The enriched feature sets provide the GNN models with more informative and discriminative representations of the molecular graphs by including detailed chemical and structural properties from the 3D optimised molecular representations. Therefore, the models can effectively learn

表5 不同分子性质预测模型的性能指标比较

数据集	模型	AUC	ACC	F1	MCC	输入特征空间
DILI	DeepDILI [4]	0.659 (不适用)	0.687 (不适用)	0.755 (不适用)	0.331 (不适用)	化学 ^b
	SSM [17]	0.691 ± 0.011	0.687 ± 0.005	0.784 ± 0.008	0.338 ± 0.030	化学 ^b
	DILIPredictor [18]	0.63 (不适用)	0.590 (N/A) [†]	0.610 (不适用)	0.160 (不适用)	化学 ^b 和生物
	DNN [19]	0.713 (不适用)	0.689 (不适用)	0.753 (不适用)	0.344 (不适用)	化学 ^b
	DNN-GATNN [20]	0.757 (不适用)	0.751 (不适用)	0.825 (不适用)	0.399 (不适用)	化学 ^{a, b}
	DILIGeNN (GraphSAGE) *	0.897 ± 0.043	0.836 ± 0.095	0.846 ± 0.106	0.698 ± 0.148	化学 ^a
临床毒理	D-MPNN [36]	0.906 ± 0.043	N/A	N/A	N/A	化学 ^{a, b}
	宝石 [37]	0.901 ± 0.013	N/A	N/A	N/A	化学 ^{a, b}
	GTOT 调校 [35]	0.779 ± 0.032	N/A	N/A	N/A	化学 ^a
	DGCL [38]	0.971 ± 0.029	N/A	N/A	N/A	化学 ^{a, b}
	HDBind-MolFormer [41]	0.988 ± 0.000	N/A	N/A	N/A	化学 ^{b, c}
	DILIGeNN (GCN) *	0.918 ± 0.028	0.950 ± 0.009	0.782 ± 0.045	0.603 ± 0.075	化学 ^a
BBBP	D-MPNN [36]	0.925 ± 0.036	N/A	N/A	N/A	化学 ^{a, b}
	宝石 [37]	0.724 ± 0.004	N/A	N/A	N/A	化学 ^{a, b}
	GTOT 调校 [35]	0.715 ± 0.008	N/A	N/A	N/A	化学 ^a
	DGCL [38]	0.738 ± 0.055	N/A	N/A	N/A	化学 ^{a, b}
	HDBind-MolFormer [41]	0.992 ± 0.001	N/A	N/A	N/A	化学 ^{b, c}
	DILIGeNN (GraphSAGE) *	0.993 ± 0.006	0.973 ± 0.012	0.982 ± 0.008	0.929 ± 0.029	化学 ^a
BACE	D-MPNN [36]	0.898 ± 0.031	N/A	N/A	N/A	化学 ^{a, b}
	宝石 [37]	0.856 ± 0.011	N/A	N/A	N/A	化学 ^{a, b}
	GTOT 调校 [35]	0.853 ± 0.015	N/A	N/A	N/A	化学 ^a
	DGCL [38]	0.915 ± 0.168	N/A	N/A	N/A	化学 ^{a, b}
	HDBind-MolCLR [41]	0.824 ± 0.005	N/A	N/A	N/A	化学 ^{a, b}
	DILIGeNN (GraphSAGE) *	0.953 ± 0.020	0.911 ± 0.034	0.906 ± 0.035	0.824 ± 0.068	化学 ^a

模型按发布日期排序。加粗的指标表示最高性能，而下划线表示第二高。所有数值均表示为平均值 ± 标准差。每项研究中使用的不同化学特征以上标表示 * 我们在DILI、BBBP和BACE预测任务中表现最好的模型是GraphSAGE，而在ClinTox预测任务中表现最好的模型是GCN。

† 平衡准确率 a 分子

图 b 化学描述符 c

SMILES 嵌入

并解释与目标属性相关的潜在模式，并实现更高的性能，这从AUC评分的显著提升中可以看出。

分子性质预测中先进模型的比较分析 DILI 预测中的性能评估

表5总结了我们的模型在DILI预测任务中与近期最先进模型的性能比较。表中包含的模型要么使用FDA发布的DILI数据集（DILIGeNN、DILIPredictor、SSM、DeepDILI），要么结合了FDA发布的

在处理附加数据以将DILI分类转换为二进制标签并删除重复化合物（DNN，DNN-GATNN）后，使用LiverTox数据集生成的数据集。其他近期的最新模型由于将在后面部分讨论，因此未包含在比较中。

DILIGeNN 模型受益于我们的方法学，该方法包括分子标准化和图生成过程。分子标准化确保了数据一致性，并减少了由互变异构体变化引起的偏差和不准确性。在图生成过程中，我们利用分子优化方法生成并优化 3D 构象，使其更真实地反映分子特征，如键长，

Table 5 Comparison of the performance metrics across different molecular property prediction models

Dataset	Model	AUC	ACC	F1	MCC	Input feature space
DILI	DeepDILI [4]	0.659 (N/A)	0.687 (N/A)	0.755 (N/A)	0.331 (N/A)	Chemical ^b
	SSM [17]	0.691 ± 0.011	0.687 ± 0.005	0.784 ± 0.008	0.338 ± 0.030	Chemical ^b
	DILIPredictor [18]	0.63 (N/A)	0.590 (N/A) [†]	0.610 (N/A)	0.160 (N/A)	Chemical ^b & Biological
	DNN [19]	0.713 (N/A)	0.689 (N/A)	0.753 (N/A)	0.344 (N/A)	Chemical ^b
	DNN-GATNN [20]	0.757 (N/A)	0.751 (N/A)	0.825 (N/A)	0.399 (N/A)	Chemical ^{a, b}
	DILIGeNN (GraphSAGE)*	0.897 ± 0.043	0.836 ± 0.095	0.846 ± 0.106	0.698 ± 0.148	Chemical ^a
ClinTox	D-MPNN [36]	0.906 ± 0.043	N/A	N/A	N/A	Chemical ^{a, b}
	GEM [37]	0.901 ± 0.013	N/A	N/A	N/A	Chemical ^{a, b}
	GTOT-Tuning [35]	0.779 ± 0.032	N/A	N/A	N/A	Chemical ^a
	DGCL [38]	0.971 ± 0.029	N/A	N/A	N/A	Chemical ^{a, b}
	HDBind-MolFormer [41]	0.988 ± 0.000	N/A	N/A	N/A	Chemical ^{b, c}
	DILIGeNN (GCN)*	0.918 ± 0.028	0.950 ± 0.009	0.782 ± 0.045	0.603 ± 0.075	Chemical ^a
BBBP	D-MPNN [36]	0.925 ± 0.036	N/A	N/A	N/A	Chemical ^{a, b}
	GEM [37]	0.724 ± 0.004	N/A	N/A	N/A	Chemical ^{a, b}
	GTOT-Tuning [35]	0.715 ± 0.008	N/A	N/A	N/A	Chemical ^a
	DGCL [38]	0.738 ± 0.055	N/A	N/A	N/A	Chemical ^{a, b}
	HDBind-MolFormer [41]	0.992 ± 0.001	N/A	N/A	N/A	Chemical ^{b, c}
	DILIGeNN (GraphSAGE)*	0.993 ± 0.006	0.973 ± 0.012	0.982 ± 0.008	0.929 ± 0.029	Chemical ^a
BACE	D-MPNN [36]	0.898 ± 0.031	N/A	N/A	N/A	Chemical ^{a, b}
	GEM [37]	0.856 ± 0.011	N/A	N/A	N/A	Chemical ^{a, b}
	GTOT-Tuning [35]	0.853 ± 0.015	N/A	N/A	N/A	Chemical ^a
	DGCL [38]	0.915 ± 0.168	N/A	N/A	N/A	Chemical ^{a, b}
	HDBind-MolCLR [41]	0.824 ± 0.005	N/A	N/A	N/A	Chemical ^{a, b}
	DILIGeNN (GraphSAGE)*	0.953 ± 0.020	0.911 ± 0.034	0.906 ± 0.035	0.824 ± 0.068	Chemical ^a

Models are ordered based on publication date. Metrics in bold represent the highest performance, while those underlined indicate the second highest. All values represented as mean ± std. Different chemical features utilised in each study are superscripted

* Our best performing model for DILI, BBBP and BACE prediction tasks is GraphSAGE, and GCN is the best performing model for ClinTox prediction task.

†Balanced Accuracy

^a Molecular Graph

^b Chemical descriptors

^c SMILES embedding

and interpret the underlying patterns associated with the target properties and achieve higher performance, as evidenced by the substantial improvements in AUC scores.

Comparative analysis of advanced models in molecular property prediction *Performance evaluation in DILI prediction*

Table 5 summarises the performance of our model compared to recent state-of-the-art models in the DILI prediction task. The models included in the table either use the FDA-released DILI datasets (DILIGeNN, DILIPredictor, SSM, DeepDILI), or combine the FDA-released

datasets with LiverTox dataset after processing the additional data to convert DILI classifications into binary labels and remove duplicate compounds (DNN, DNN-GATNN). Other recent state-of-the-art models were excluded from the comparison, as discussed in a later section.

DILIGeNN models benefits from our methodology that includes molecular standardisation and graph generation process. The molecular standardisation ensures data consistency and reduces biases and inaccuracies from tautomeric variability. During the graph generation, we utilised molecular optimisation methods to generate and optimise 3D conformers to a more realistic representation to capture molecular features such as bond lengths,

部分电荷和准确的几何构型。这些特征对于准确预测药物性肝损伤 (DILI) 结果至关重要, 因为它们提供了关于分子空间结构和电子分布的详细信息。

我们的 DILIGeNN 模型 (GraphSAGE) 在表中优于所有其他模型, 取得了最高的 AUC 值 0.897, 超过了 DNN-GATNN 的 AUC 值 0.757。这证明了我们的模型能够有效区分 DILI 阳性和 DILI 阴性化合物, 利用其图卷积结构捕捉复杂的分子特征。

与使用二维分子图特征以及额外子结构指纹 (MACCS) 来训练DNN-GAT集成模型的 DNN-GATNN集成模型不同, [20], DILIGeNN模型在 GNN架构中只使用单一的分子图表示。尽管采用单一 GNN架构且没有额外的化学指纹特征, 我们的方法通过结合从分子优化和模型微调步骤中衍生的空间和静电特征, 仍然是最有效的。

此外, DILIGeNN 模型不仅超越了仅基于化学数据的其他模型, 还表现优于 DILIPredictor, 后者使用化学数据以及预测的体内和体外生物数据实现了 0.79 的 AUC-PR 和 0.63 的 AUC-ROC。来自 DILIPredictor 和其他研究的证据表明, 整合生物特征并不总是能提高 DILI 模型的预测性能 [18, 64]。

这很可能反映了 DILI 的多因素性质, 其涉及复杂且理解不完全的机制, 而这些机制难以通过现有的生物标志物捕捉 [2]。尽管如此, DILIPredictor 采用的方法代表了整合化学和生物数据的重要一步, 可提供机制性见解并提高模型预测的可解释性。

通过专注于化学特征, 我们的基于图的架构能够有效识别与DILI预测相关的关键分子特征。这在早期药物开发和药物再利用阶段特别有利, 因为此时DILI相关的生物数据通常不可用。虽然我们的模型仅关注化学特征, 但生物数据在解释模型预测背后的生物过程方面仍然起着重要作用。随着对可靠生物标志物理解的提升和可用性的增加, 未来的模型可以更好地整合生物特征, 从而提升预测性能并改善对DILI发病机制的理解。

在Clintox、BBBP和BACE数据集上的性能评估 表5展示了我们表现最佳的基于图的模型在Clintox、BBBP和BACE分子性质预测任务中与最新最先进模型的性能对比。这些分子性质预测任务对于评估化合物的潜在毒性、渗透性和生化活性至关重要。

DILIGeNN 和 GTOT-tuning [35] 是唯一利用单一分子图表示而无需额外化学描述符或分子嵌入来进行预测任务的方法, 并且两者都采用了微调技术。所有其他方法要么通过化学描述符 [36–38], 进行补充, 要么在 HDBind-MolFormer [41] 的情况下, 将化学描述符与 SMILES 嵌入结合语言模型使用。

我们的模型 DILIGeNN (GraphSAGE) 在 BBBP 和 BACE 数据集上分别实现了 0.993 和 0.953 的 AUC 分数, 分别超过了之前的最先进水平, 其中 BBBP 数据集的 HDBind-MolFormer [41] 为 0.992, BACE 数据集的 DGCL [38] 为 0.915。在 ClinTox 数据集上, 我们的 DILIGeNN (GCN) 模型的 AUC 分数为 0.918, 排名第三, 仅次于 HDBind-MolFormer 和 DGCL。

这些结果突出显示了我们的模型在从分子图捕捉分子特征以进行渗透性和生化活性预测以及DILI预测方面的能力。这强调了将先进的化学特征工程与GNN架构相结合以改进分子预测任务的重要性。

与其他方法的比较及局限性

准确的评估方法对于评估模型的性能和可推广性至关重要, 尤其是在药物应用如DILI预测中。Lim等人[17] 和 Li等人[4] 的先前研究使用了DeepDILI数据集中的预定义训练集和测试集, 而没有采用交叉验证来评估整个数据集。Li等人进一步将他们的训练数据分为两个子集, 以优化其基础分类器的超参数, 而Lim等人则使用 Therapeutics Data Commons的DILI基准数据集[65] 作为外部验证数据集[17]。然而, 该数据集包含来自与训练数据集 (FDA DILIST数据集) 相同来源的药物, 使用重叠数据会削弱外部验证的目的。

相比之下, 我们的研究采用嵌套交叉验证框架对 GNN在DILI数据集以及其他分子性质预测数据集 (包括Clintox、BBBP和BACE) 上的训练和评估。我们的方法使用内部训练循环, 允许数据集中的每个实例用于训练和验证。

partial charges, and accurate geometric configurations. These features are crucial for accurately predicting DILI outcomes, as they provide detailed information about the molecule's spatial structure and electron distribution.

Our DILIGeNN model (GraphSAGE) outperforms all other models in the table, achieving the highest AUC of 0.897, surpassing DNN-GATNN's AUC of 0.757. This demonstrates our model's ability to effectively distinguish between DILI-positive and DILI-negative compounds, leveraging its graph convolutional architecture to capture complex molecular features.

Unlike the DNN-GATNN ensemble model, which uses 2D molecular graph features together with an additional substructure fingerprint (MACCS) to train DNN-GAT ensemble model [20], DILIGeNN model utilises a single molecular graph representation within a GNN architecture. Despite employing a single GNN architecture without additional chemical fingerprint features, our approach is the most effective by incorporating spatial and electrostatic features in derived from molecular optimisation and model fine-tuning steps.

In addition, the DILIGeNN model not only surpasses other models based solely on chemical data but also outperforms DILIPredictor, which achieves an AUC-PR of 0.79 and AUC-ROC of 0.63 using both chemical and predicted *in vivo* and *in vitro* biological data. Evidence from both DILIPredictor and the other study indicates that incorporating biological features does not consistently improve predictive performance for DILI models [18, 64]. This likely reflects the multifactorial nature of DILI, which involves complex and incompletely understood mechanisms that are difficult to capture through current biological markers [2]. Nevertheless, the approach taken by DILIPredictor represents an important step in integrating chemical and biological data to provide mechanistic insight and improve the interpretability of the model predictions.

By focusing exclusively on chemical features, our graph-based architecture effectively identifies key molecular characteristics associated with DILI prediction. This is particularly advantageous in the early-stage drug development and drug-repurposing phase where DILI-related biological data is often unavailable. While our model focuses on chemical features alone, biological data still play an important role in interpreting the biological processes underlying a model's prediction. With improved understanding and availability of reliable biological markers, future models could better integrate biological features to enhance predictive performance and improve mechanistic understanding of DILI pathogenesis.

Performance evaluation on Clintox, BBBP, and BACE datasets

Table 5 illustrates the performance of our top-performing graph-based models in comparison to the recent state-of-the-art models for the Clintox, BBBP and BACE molecular property prediction tasks. These molecular property prediction tasks are crucial in evaluating a compound's potential toxicity, permeability and biochemical activity.

DILIGeNN and GTOT-tuning [35] are the only methods that utilise a single molecular graph representation without additional chemical descriptors or molecular embeddings for the prediction tasks and both leverages fine-tuning techniques. All other methods either complement this with chemical descriptors [36–38], or in the case of HDBind-MolFormer [41] combining chemical descriptors with SMILES embedding with language models.

Our model, DILIGeNN (GraphSAGE), achieved AUC scores of 0.993 and 0.953 on the BBBP and BACE datasets, respectively, outperforming the previous state-of-the-art AUC of 0.992 by HDBind-MolFormer [41] for the BBBP and 0.915 by DGCL [38] for the BACE datasets. For the ClinTox dataset, our DILIGeNN (GCN) model with an AUC score of 0.918 ranks third behind HDBind-MolFormer and DGCL.

These results highlight the our model's capability in capturing molecular features from the molecular graph for permeability and biochemical activity prediction as well as the DILI prediction. This emphasises the importance of integrating advanced chemical feature engineering with the GNN architectures for improved molecular prediction tasks.

Comparison with other approaches and limitations

Accurate evaluation methodologies are crucial for assessing the performance and generalisability of models, especially in medicinal applications such as DILI prediction. Previous studies by Lim et al., [17] and Li et al., [4] used predefined training and test sets from the DeepDILI dataset, without employing cross-validation to evaluate the entire dataset. Li et al., further divided their training data into two subsets to optimise hyperparameters for their base classifiers, where as Lim et al. used the Therapeutics Data Commons' DILI benchmark dataset [65] as an external validation dataset [17]. However, this dataset contains drugs originating from the same source as the training dataset (FDA DILIST dataset), and using overlapping data defeats the purpose of the external validation.

In contrast, our study employed the nested CV framework for training and evaluating the GNNs on the DILI dataset, as well as on other molecular property prediction datasets, including Clintox, BBBP, and BACE. Our method uses inner training loops, allowing each instance in the dataset to be used for both training and validation

集合。此外, 在当前最先进的DILI预测模型中, 我们的方法是唯一一个在外部测试拆分中评估整个数据集的模型, 从而提供对模型性能和泛化能力的更可靠评估。

还有其他DILI预测模型报告的性能指标比本研究评估的更高 [66, 67]。然而, 它们的可靠性可能受到使用重叠DILI数据集和不同DILI分类标准的限制。例如, DILIST是DILIRank数据集的增强版本, 通过整合其他DILI来源开发而成

[10, 11, 43–45]。我们承认, 本研究使用的基准数据集可能存在局限性, 包括化学表示不一致和分子结构无效的问题 [71]。为了解决这些问题, 我们采用了自己的分子标准化方法, 以提高结构一致性和未来的可重复性。本研究使用的标准化方法的代码, 以及原始和标准化的 SMILES 字符串, 都可以在我们的 GitHub 上获取。这些研究整合了重叠的DILI资源数据, 但未提供充分的细节说明用于识别和移除重复化合物或将不同DILI分类转换为二元标签的方法。这种数据整合可能引入DILI标注的变异性, 并增加数据泄露的风险, 从而夸大模型性能指标。相比之下, 我们的研究使用了FDA最新的、系统地整理的DILIST数据集, 为模型开发提供了一致且统一的基准。

我们的方法存在一些局限性。在DILI预测研究中, 我们只使用了化学特征。生物特征也可用, 并且如其他方法所示, 可能提供更好的模型可解释性 [18, 64]。尽管我们的方法在ClinTox数据集上未达到最高性能, 但表现更好的两种方法使用了本研究中未包含的额外化学特征。目前仍需研究添加这些特征是否能进一步提升DILIGeNN的性能。尽管GNN本质上是可解释的, 在本研究中我们并未专注于提取模型决策过程的任何信息, 而是集中于分类性能。最近关于异构信息网络(一种基于图的方法)的工作表明, 整合生物网络信息, 例如药物-蛋白和蛋白-疾病关联, 可以通过揭示调控通路提高模型的可解释性。

在ClinTox、BBBP和BACE预测研究中, 我们无法用所有指标将我们的模型与其他方法进行比较, 因为这些指标并非全部可用, 因此我们在报告中包含了多个指标, 以便他人能够

在未来, 我们将与 DILIGeNN 进行比较, DILIGeNN 使用了多个测试集划分。此外, 增强特征源自力场计算 [51], 可能无法反映输入分子在与参与生物反应的蛋白质的分子间相互作用中的真实分子表示。然而, 我们的研究表明, 与基于二维分子表示的传统分子图相比, 从三维优化结构中引入增强特征能够改善对输入分子的表征。

我们承认, 本研究使用的基准数据集可能存在局限性, 包括化学表示不一致和分子结构无效的问题 [71]。为了解决这些问题, 我们采用了自己的分子标准化方法, 以提高结构一致性和未来的可重复性。本研究使用的标准化方法的代码, 以及原始和标准化的 SMILES 字符串, 都可以在我们的 GitHub 上获取。

结论

在本研究中, 我们开发并评估了四种主要的图神经网络(GNN)架构, 用于使用定制设计的图特征作为化学输入来预测药物性肝损伤(DILI)及其他分子性质。我们的方法(DILIGeNN)在DILI、血脑屏障通透性(BBBP)、BACE预测任务上显示出比现有最先进方法更好的性能。DILI预测中表现最好的模型实现了0.897的AUC(GraphSAGE), 超过了利用额外分子指纹和图结构数据的DNN-GATNN(AUC 0.757)。此外, 我们在BBBP和BACE任务中表现最好的模型(均为GraphSAGE)也超过了最先进的模型, 推动了跨多样化化学空间的分子性质预测。这些结果突显了DILIGeNN模型在使用更现实的化学特征捕捉复杂分子特征方面的有效性, 这对于分子性质的准确预测至关重要。此外, 我们的嵌套交叉验证(nested CV)框架提高了预测性能。

我们方法的一个关键创新是使用定制设计的图特征, 这显著增强了模型表示和学习分子数据的能力。通过引入更真实的分子特征, 如键长和部分电荷, 我们提高了DILIGeNN模型在不同分子预测任务中的性能。我们定制的特征工程对于实现比现有模型更好的性能至关重要。未来的工作将涉及引入可解释性。

sets. Furthermore, our approach is the only one among current state-of-the-art DILI prediction models that evaluates the entire dataset across outer test splits, providing more reliable assessment of the model's performance and generalisability.

There are other DILI prediction models that have reported higher performance metrics than those evaluated in this study [66, 67]. However, their reliability may be limited by the use of overlapping DILI datasets and different DILI classification standards. For instance, DILIST is an augmented version of the DILIRank dataset, developed by incorporating other DILI sources [10, 11, 43–45]. These studies integrated data from the overlapping DILI resources without providing sufficient detail on the methods used to identify and remove duplicate compounds or to convert different DILI classifications into a binary label. Such data integration can introduce variability in DILI annotation and increase the risk of data leakage, thereby inflating model performance metrics. In contrast, our study used the FDA's latest and systematically curated DILIST dataset, providing a consistent and uniform benchmark for model development.

Our approaches have some limitations. In the DILI prediction study, we only make use of chemical features. Biological features are also available and may provide better interpretability of the model as shown with the other approaches [18, 64]. While our method did not achieve the highest performance on the ClinTox dataset, the two methods with better performance used additional chemical features not included in this work. It remains to be investigated whether adding these features could further boost performance for DILIGeNN. While GNNs are explainable by nature, in this work we have not focussed on extracting any information about the decision making process of the models and focussed on classification performance. Recent works on heterogeneous information networks, a type of graph-based method, demonstrate that integrating biological network information, such as drug-protein and protein-disease associations, can improve model interpretability by revealing regulatory pathways and connectivity patterns that underlie drugs' mechanisms of action [68, 69]. Moreover, generative modelling approaches have not been explored in this study, such as those applied in AMP-Designer to design molecular structures with desirable molecular properties [70]. Similar techniques could be adapted in the future to generate small molecules with reduced DILI risk by training and fine-tuning the model with the DILI dataset.

In ClinTox, BBBP and the BACE prediction study, we could not compare our models with other methods for all metrics as these are not all available, hence we included multiple metrics in our reporting to allow others to

compare against DILIGeNN for which we used multiple test set splits in future. In addition, the augmented features are derived from the force field calculations [51] and may not reflect the true molecular representation of the input molecules in their intermolecular interactions with proteins involved in the biological response. However, our study demonstrates that incorporating the augmented features from the 3D optimised structures improves the characterisation of the input molecules compared to traditional molecular graphs from the 2D molecular representation.

We acknowledge that benchmark datasets used in this study may have limitations, including issues with inconsistent chemical representations and invalid molecular structures [71]. To address these concerns, we applied our own molecular standardisation method to improve structural consistency and future reproducibility. The code for our standardisation method, as well as both the original and standardised SMILES strings used in this study, is available in our [GitHub](#).

Conclusion

In this study, we developed and evaluated four prominent GNN architectures for predicting DILI and other molecular properties using custom-designed graph features as chemical inputs. Our method (DILIGeNN) demonstrated improved performance in comparison to the state-of-the-art methods for DILI, BBBP, BACE prediction tasks. The best performing model in DILI prediction achieved an AUC of 0.897 (GraphSAGE) surpassing DNN-GATNN (AUC = 0.757) that utilises an additional molecular fingerprint with graph-structured data. Additionally, our best performing models in BBBP and BACE (both GraphSAGE) outperformed the state-of-the-art models, advancing molecular property prediction across diverse chemical spaces. These results highlights the effectiveness of DILIGeNN models in capturing complex molecular features with more realistic chemical features, which is crucial for accurate predictions of molecular properties. In addition, our nested CV framework improved predictive performance of GNN models over multiple stages of optimisation and fine-tuning, without losing generalisability.

A key innovation in our approach is the use of the custom-designed graph features, which significantly enhanced the models' ability to represent and learn from molecular data. By incorporating more realistic molecular features such as bond lengths and partial charges, we have improved performance of DILIGeNN models across different molecular prediction tasks. Our custom feature engineering was crucial for achieving improved performance over current models. Future work involves incorporating an explainability

用于提取每个分子性质预测任务相关子图结构的函数。

补充信息

在线版本包含可在 <https://doi.org/10.1186/s13321-025-01068-3> 获取的补充材料。

补充材料 1。

作者贡献

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数据可用性

所有代码和数据都提供在 https://github.com/tlee23-ic/GNN_DILI。

声明

利益冲突

The authors declare no competing interests.

作者详情 1

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function to extract relevant subgraph structures for each molecular property prediction task.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13321-025-01068-3>.

Supplementary material 1.

Author contributions

Conceptualisation, T.L. and J.M.P.; Methodology, T.L.; Formal Analysis and Investigation, T.L.; Supervision, J.M.P.; Writing – Original Draft: T.L.; Writing – Review & Editing: T.L. and J.M.P. Both authors have read and agreed to the published version of the manuscript.

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Data availability

All code and data are provided at https://github.com/tlee23-ic/GNN_DILI.

Declarations

Competing interests

The authors declare no Competing interests.

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